

INDUSTRIAL NETWORKING
MENG 4010

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4. Internet layer protocols

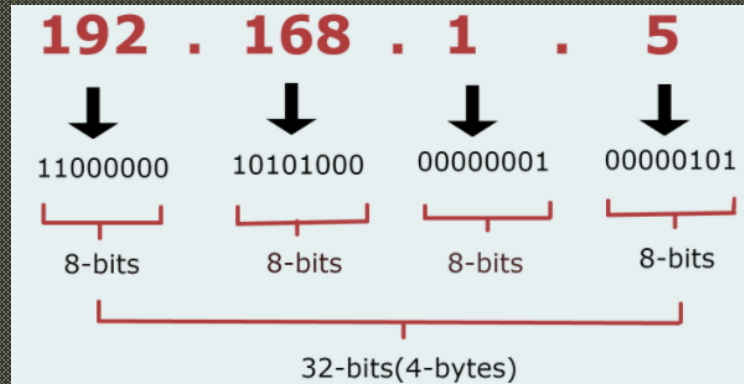
4.1. IPv4 Addressing

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4.1. IPv4 Addressing

- The Internet Protocol (IP) is responsible for routing packets towards their destination.
- The routing is performed based on the IP addresses, embedded in the header attached to each packet forwarded by IP.
- Definition:
 - An IPv4 address is a 32-bit number divided into four parts, called octets.
 - Each octet has a value from 0 to 255.
- Format:
 - Written in dotted decimal notation: four decimal numbers separated by periods.
 - Example:
 - 192.168.1.5
 - 192 = 1st octet
 - 168 = 2nd octet
 - 1 = 3rd octet
 - 5 = 4th octet



The IPv4 address consists of 32 bits (11000000 10101000 00000001 00000101).
Each octet is converted to its decimal equivalent (192. 168. 1. 5).

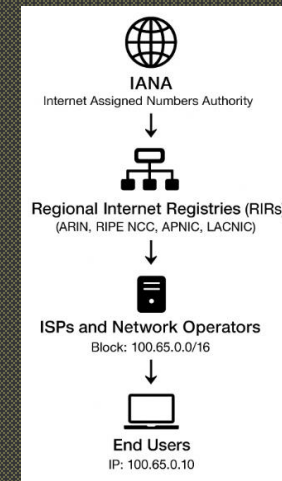
4.1. Internet Protocol Version 4 (IPv4)

IP Addresses (Hierarchy)

1. IANA (Internet Assigned Numbers Authority)
 - The global authority that manages the entire pool of IP addresses.
 - Allocates large blocks of IP addresses to five Regional Internet Registries (RIRs).
2. Regional Internet Registries (RIRs)
 - Responsible for distributing IP address blocks within their regions.
 - Each RIR then gives smaller blocks to ISPs, governments, and large organizations.
3. ISPs (Internet Service Providers) and Network Operators
 - ISPs receive IP address blocks from their RIR.
 - They assign IP addresses to customers for internet access.
 - Example: When you connect your home router to the internet, your ISP gives it a public IP address.
4. End Users (You & Organizations)
 - Home Users: Your ISP assigns a public IP address to your home router.
 - Companies/Universities: A network administrator assigns IP addresses to internal devices (servers, printers, PCs) using private IP ranges.

Examples of RIRs:

- ARIN – North America
- RIPE NCC – Europe, Middle East, Central Asia
- APNIC – Asia-Pacific
- LACNIC – Latin America & Caribbean
- AFRINIC – Africa



4.1. Internet Protocol Version 4 (IPv4)

IP Addresses (Hierarchy)

- Dynamic IP Allocation (Most Common for Home Users)
 - Your ISP gives you an IP address automatically when you connect.
 - The address can change every time your router restarts.
 - Example: Today your public IP might be 64.32.10.15, but tomorrow it could be 64.32.10.88.
- Static IP Allocation (Permanent)
 - Businesses can request static IPs that do not change.
 - Useful for:
 - Hosting a website
 - Running a mail server
 - Remote access to security cameras or VPN servers
- Internal (Private) Networks & DHCP
 - Inside your home or office network, devices get private IP addresses like:
 - 192.168.1.2 or 192.168.1.10 (PCs, phones, printers, etc.)
 - The router uses DHCP (Dynamic Host Configuration Protocol) to automatically assign these IPs so devices can communicate with each other and access the internet.

4.1. Internet Protocol Version 4 (IPv4)

Network ID and Host ID

- The MAC address is a unique identifier assigned to each network device (e.g., a NIC) at the time of manufacture. A good analogy is a human ID or Social Security number - a permanent identifier that is unique to each person.
- On a local network (LAN), the MAC address is sufficient for identifying and communicating between devices, since all devices can directly “see” each other.
- However, when communication extends beyond the local network, such as sending data across the internet, the MAC address alone is not enough. Just like a letter must first travel through the postal system to reach another country before the recipient’s ID becomes useful, data requires a location address.
- In TCP/IP networks, this location is provided by the IP address. Unlike the MAC address, the IP address is not permanently tied to the device; instead, it tells the network where the device is currently located.

Consider the address:

205 Humber College Blvd.
Etobicoke, M9W 5L7
Ontario, Canada

- The first part, 205 Humber College Blvd., enables the local postal deliveryman at the post office in Etobicoke, M9W 5L7 to deliver a letter to this specific residence.
- The second part, Ontario Canada, enables the International Postal System to route the letter towards its destination post office from anywhere in the world.

4.1. Internet Protocol Version 4 (IPv4)

Network ID and Host ID

In similar fashion, An IP address is made up of two distinct parts:

- Network ID (NetID)
 - Identifies the specific network the device belongs to.
 - Allows routers on the internet to forward packets toward the correct network, no matter where the packet originates in the world.
 - Think of it like the street name in a mailing address, it ensures the letter reaches the right neighborhood before looking for the exact house.
- Host ID (HostID)
 - Identifies the specific device (host) within that network.
 - Used by the router inside the destination network to deliver the packet to the correct computer, printer, or device.
 - Think of it like the house number, it ensures the packet is delivered to the correct device on that network.

Example 1

- IP Address: 142.214.143.10
- Subnet Mask: 255.255.255.0
 - This means the first three octets (142.214.143) represent the Network ID.
 - The last octet (10) represents the Host ID.
- So:
- Network ID (NetID): 142.214.143.0 → The network this device belongs to
- Host ID (HostID): 10 → The specific device on that network

Another Example

- IP Address: 192.168.1.25
- Subnet Mask: 255.255.255.0
 - Network ID: 192.168.1.0 (the entire network)
 - Host ID: 25 (the specific device)
- If another computer had IP 192.168.1.50, it would be on the same network (192.168.1.0), but a different host (50).

4.1. Internet Protocol Version 4 (IPv4)

Network ID and Host ID

- Network ID & Host ID
 - Network ID: Identifies the network the computer belongs to.
 - Host ID: Identifies the specific device (host) within that network.
 - Subnet Mask: Determines which part of the IP address is the network ID and which part is the host ID.
- Subnet Mask
 - Definition:
 - A 32-bit number (like an IP address) with binary 1s followed by binary 0s.
 - Each 255 in the subnet mask = 8 binary 1s (11111111).
 - Examples of Common Subnet Masks:
 - 255.0.0.0 → Network ID = first octet only
 - 255.255.0.0 → Network ID = first two octets
 - 255.255.255.0 → Network ID = first three octets

The following is not allowed for a Host ID:

- .0 → Reserved for network ID (used to represent the entire network, not a single device)
- .255 (last address) → Reserved for broadcast (used to send messages to every device in that network)

4.1. IPv4 Addressing

Subnet Masks and Binary Math Determine Network ID

- Digital devices use binary math to figure out the network portion of the IP address.
- This is done with a logical AND operation (think of it like multiplying 0s and 1s).
- Subnet masks define which bits are network vs which bits are host.
- Logical AND Operation
 - Works like binary multiplication:
 - $0 \text{ AND } 0 = 0$
 - $1 \text{ AND } 0 = 0$
 - $0 \text{ AND } 1 = 0$
 - $1 \text{ AND } 1 = 1$
 - This means:
 - If there is a 1 in the subnet mask $\rightarrow$ keep the bit from the IP address.
 - If there is a 0 in the subnet mask $\rightarrow$ turn that bit into 0.

Rule of Thumb

- 1s in subnet mask = network ID bits
- 0s in subnet mask = host ID bits

- Example 1: Finding Network ID
- IP Address: 172.31.100.6
10101100.00011111.01100100.00000110
- Subnet Mask: 255.255.0.0
11111111.11111111.00000000.00000000
- Logical AND:
10101100.00011111.00000000.00000000
- Result in decimal: 172.31.0.0 (network ID)

Convert 125 to binary using subtraction method:

- $125 < 128 \rightarrow 0$ under 128 $\rightarrow$ (test number stays 125)
- $125 \geq 64 \rightarrow 1$ under 64 $\rightarrow 125 - 64 = 61$
- $61 \geq 32 \rightarrow 1$ under 32 $\rightarrow 61 - 32 = 29$
- $29 \geq 16 \rightarrow 1$ under 16 $\rightarrow 29 - 16 = 13$
- $13 \geq 8 \rightarrow 1$ under 8 $\rightarrow 13 - 8 = 5$
- $5 \geq 4 \rightarrow 1$ under 4 $\rightarrow 5 - 4 = 1$
- $1 < 2 \rightarrow 0$ under 2
- $1 \geq 1 \rightarrow 1$ under 1 $\rightarrow 1 - 1 = 0$ (done)
- Result: $125 = 01111101$ (binary)

Convert 11010011 to decimal:

- Write down decimal values where there is a 1:
 $128 + 64 + 16 + 2 + 1 = 211$
- So, $11010011 = 211$

- Example 2: Finding Network ID
- IP Address: 172.16.67.5
10101100.00010000.01000011.00000101
- Subnet Mask: 255.255.192.0
11111111.11111111.11000000.00000000
- Logical AND:
10101100.00010000.01000000.00000000
- Result in decimal:
172.16.64.0 (network ID)
- Remaining bits = host ID
 $00000011.00000101 \rightarrow 3.5$

4.1. IPv4 Addressing

Subnet Masks and Binary Math Determine Network ID

Example 1

- IP Address: 10.1.221.101
- Subnet Mask: 255.0.0.0
- Analysis:
 - First octet is network ID (255 = 8 ones → first octet is network).
 - Network ID: 10.0.0.0
 - Host ID: 1.221.101
- Meaning: Computer belongs to network 10.0.0.0, and its unique host is 1.221.101.

Example 2

- IP Address: 172.31.100.6
- Subnet Mask: 255.255.0.0
- Analysis:
 - First two octets are network ID (16 ones).
 - Network ID: 172.31.0.0
 - Host ID: 100.6
- Meaning: Device is on network 172.31.0.0, host 100.6.

Example 3

- IP Address: 192.168.14.250
- Subnet Mask: 255.255.255.0
- Analysis:
 - First three octets are network ID (24 ones).
 - Network ID: 192.168.14.0
 - Host ID: 250
- Meaning: Device is on network 192.168.14.0, host 250.

4.1. IPv4 Addressing

Subnet Masks and Binary Math Determine Network ID

Example with Partial Octet (Advanced)

- IP Address: 172.16.67.5
- Subnet Mask: 255.255.192.0
- Analysis:
 - Third octet is partially used (first 2 bits are 1s, rest 6 are 0s).
 - Binary mask: 11111111.11111111.11000000.00000000
 - Network ID: 172.16.64.0 (because 64 is where the binary pattern starts)
 - Host ID: 3.5 (because $67 - 64 = 3$, so host portion is 3.5)
- Meaning: The network covers IPs from 172.16.64.0 to 172.16.127.255

1. The third octet of the subnet mask is 192.

- Convert 192 to binary: 11000000
- This means 2 network bits + 6 host bits
- Block size = $256 - 192 = 64$
- This tells us:
- Networks will start every 64 numbers in the third octet.
- Network blocks are:
 - 0 – 63
 - 64 – 127
 - 128 – 191
 - 192 – 255
- Our third octet is 67 → falls in the 64 – 127 block, so network ID = 172.16.64.0

2. The broadcast address is the last address in the network block.

- Network starts at 64
- Add the block size (64) → $64 + 64 = 128$
- The next block would start at 128, so we go one number back to get the broadcast:
 - $128 - 1 = 127$
- This means:
 - Broadcast address = 172.16.127.255

3. Hosts can only use addresses between the network ID and the broadcast address.

- First valid host = network ID + 1
 - Network ID = 172.16.64.0
 - First valid = 172.16.64.1
- Last valid host = broadcast address – 1
 - Broadcast = 172.16.127.255
 - Last valid = 172.16.127.254

Item	Address
Network ID	172.16.64.0
First Host	172.16.64.1
Last Host	172.16.127.254
Broadcast Address	172.16.127.255

.0 → Reserved for network ID (used to represent the entire network, not a single device)
.255 (last address) → Reserved for broadcast (used to send messages to every device in that network)
So, the valid host range is everything between those two addresses: 172.16.64.1 → 172.16.127.254

4.1. IPv4 ADDRESSES

IP Address Classes

IP Address Classes Overview

- IPv4 addresses are grouped into Classes A, B, C, D, and E based on the first octet (first number in the IP address).
- Classes A, B, and C are for network devices (hosts).
- Classes D and E are special-use (multicast & experimental).

Class	7	6	5	4	3	2	1	0	7	6	5	4	3	2	1	0	7	6	5	4	3	2	1	0	7	6	5	4	3	2	1	0									
A	0	Network ID							Host ID																																
B	1	0	Network ID														Host ID																								
C	1	1	0	Network ID																		Host ID																			
D	1	1	1	0	Multicast Address																																				
E	1	1	1	1	Reserved for future use																																				
IP Address Structure																																									

4.1. IPv4 ADDRESSES

IP Address Classes

Although modern networks use classless addressing (CIDR), the legacy classful system remains fundamental to understanding IP design. It categorized networks into three types:

- Class A: Few networks, each with a vast number of hosts.
- Class B: A moderate number of networks, each with a moderate number of hosts.
- Class C: Many networks, each with a limited number of hosts.
- For class A, the first bit is fixed as '0'
- For class B the first 2 bits are fixed as '10'
- For class C the first 3 bits are fixed as '110'
- For class D the first 4 bits are fixed as '1110'
- For class E the first 5 bits are fixed as '11110'

Class	7	6	5	4	3	2	1	0	7	6	5	4	3	2	1	0	7	6	5	4	3	2	1	0	7	6	5	4	3	2	1	0		
A	0	Network ID							Host ID																									
B	1	0	Network ID															Host ID																
C	1	1	0	Network ID																				Host ID										
D	1	1	1	0	Multicast Address																													
E	1	1	1	1	Reserved for future use																													
IP Address Structure																																		

4.1. IPv4 ADDRESSES

IP Address Classes

Class A

- First Octet Range: 1 – 126 (0 and 127 are reserved); In binary: 00000001_2 (1_{10}) and 01111110_2 (126_{10})
- Default Subnet Mask: 255.0.0.0
- Network ID: First 8 bits (first octet)
- Host ID: Remaining 24 bits (last three octets)
- Max Hosts: 16 777 214 (per network)
- Example:
 - IP Address: 10.159.44.201
 - Network Address: 10.0.0.0
 - First Host: 10.0.0.1
 - Last Host: 10.255.255.254
- Use Case: Large organizations, ISPs, governments.
- Note:
 - 127.x.x.x range is reserved for loopback testing (refers to the local machine).
 - 127.0.0.1 is called localhost – reserved name for local machine.

Class	7	6	5	4	3	2	1	0	7	6	5	4	3	2	1	0	7	6	5	4	3	2	1	0	7	6	5	4	3	2	1	0				
A	0	Network ID							Host ID																											
B	1	0	Network ID													Host ID																				
C	1	1	0	Network ID																		Host ID														
D	1	1	1	0	Multicast Address																															
E	1	1	1	1	Reserved for future use																															
IP Address Structure																																				

First Octet / Class	7	6	5	4	3	2	1	0
Lowest in Binary	0	0	0	0	0	0	0	1
Lowest in Decimal	1							
Highest in Binary	0	1	1	1	1	1	1	0
Highest in Decimal	126							

$$2^7 = 128;$$

- 0 and 127 reserved / not allowed
- Remaining: 1 - 126

$$2^{24} = 16\,777\,216;$$

- 0 and 255 reserved / not allowed
- Remaining: 1 – 16 777 214

4.1. IPv4 ADDRESSES

IP Address Classes

Class B

- First Octet Range: 128 – 191; In binary: 10000000_2 (128_{10}) and 10111111_2 (191_{10})
- Default Subnet Mask: 255.255.0.0
- Network ID: First 16 bits (first two octets)
- Host ID: Last two octets
- Max Hosts: 65,534 (per network)
- Example:
 - IP Address: 172.17.11.4
 - Network Address: 172.17.0.0
 - First Host: 172.17.0.1
 - Last Host: 172.17.255.254
- Use Case: Medium-to-large networks (universities, companies).

Class	7	6	5	4	3	2	1	0	7	6	5	4	3	2	1	0	7	6	5	4	3	2	1	0	7	6	5	4	3	2	1	0				
A	0	Network ID							Host ID																											
B	1	0	Network ID													Host ID																				
C	1	1	0	Network ID																		Host ID														
D	1	1	1	0	Multicast Address																															
E	1	1	1	1	Reserved for future use																															
IP Address Structure																																				

First Octet / Class	7	6	5	4	3	2	1	0
Lowest in Binary	1	0	0	0	0	0	0	0
Lowest in Decimal	128							
Highest in Binary	1	0	1	1	1	1	1	1
Highest in Decimal	191							

$$2^{14} = 16\,384$$

- Network IDs: 16 384

$$2^{16} = 65\,536$$

- 0 and 255 reserved / not allowed
- Hosts: 65 534

4.1. IPv4 ADDRESSES

IP Address Classes

First Octet / Class	7	6	5	4	3	2	1	0
Lowest in Binary	1	1	0	0	0	0	0	0
Lowest in Decimal	192							
Highest in Binary	1	1	0	1	1	1	1	1
Highest in Decimal	223							

Class C

- First Octet Range: 192 – 223; In binary: 11000000_2 (192_{10}) and 11011111_2 (223_{10})
- Default Subnet Mask: 255.255.255.0
- Network ID: First 24 bits (first three octets)
- Host ID: Last octet
- Max Hosts: 254 (per network)
- Example:
 - IP Address: 203.0.113.254
 - Network Address: 203.0.113.0
 - First Host: 203.0.113.1
 - Last Host: 203.0.113.254
- Use Case: Small networks (offices, home networks).

Class	7	6	5	4	3	2	1	0	7	6	5	4	3	2	1	0	7	6	5	4	3	2	1	0
A	0	Network ID							Host ID															
B	1	0	Network ID										Host ID											
C	1	1	0	Network ID														Host ID						
D	1	1	1	0	Multicast Address																			
E	1	1	1	1	Reserved for future use																			
IP Address Structure																								

$$2^{21} = 2\,097\,152$$

- Network IDs: 2 097 152

$$2^8 = 256$$

- 0 and 255 reserved / not allowed
- Hosts: 254

Class	Range
A	1.x.y.z to 126.x.y.z
B	128.x.y.z to 191.x.y.z
C	192.x.y.z to 223.x.y.z

Class	Number of Networks	Hosts per Network
A	126	16 777 216
B	16 384	65 534
C	2 097 152	254

4.1. IPv4 ADDRESSES

IP Address Classes

Class D

- First Octet Range: 224 - 239
- Purpose: Multicasting (sending data to multiple devices at once)
- Examples of Use: Video streaming, online gaming, conferencing.
- Note: Cannot assign to individual hosts

First Octet / Class	7	6	5	4	3	2	1	0
Lowest in Binary	1	1	1	0	0	0	0	0
Lowest in Decimal	224							
Highest in Binary	1	1	1	0	1	1	1	1
Highest in Decimal	239							

Class E

- First Octet Range: 240 - 255
- Purpose: Experimental use
- Note: Not used for hosts

Class E is using all the range of the remaining Network IDs: 240 - 255

Class	7	6	5	4	3	2	1	0	7	6	5	4	3	2	1	0	7	6	5	4	3	2	1	0	7	6	5	4	3	2	1	0
A	0	Network ID							Host ID																							
B	1	0	Network ID														Host ID															
C	1	1	0	Network ID																		Host ID										
D	1	1	1	0	Multicast Address																											
E	1	1	1	1	Reserved for future use																											

IP Address Structure

- Two addresses per network are always reserved:
 - Network address (all host bits = 0) → identifies the network
 - Broadcast address (all host bits = 1) → used to send data to all hosts on that network
 - Example for Class C: 203.0.113.0 (network), 203.0.113.255 (broadcast)
- CIDR (Classless Inter-Domain Routing):
 - Replaced rigid classes with flexible prefix lengths (e.g., /24 instead of "Class C")
 - Example: 192.168.1.1/24 means first 24 bits are network ID → equivalent to 255.255.255.0

4.1. IPv4 ADDRESSES

Private IP Addresses

- Private IP addresses are used inside local networks (home, office) and cannot be accessed directly from the Internet.
- These are the addresses you usually see on home Wi-Fi routers.
 - Example: 192.168.1.1 is a common default gateway address.

Class	Range	Number of Networks	Example Addresses
Class A	10.0.0.0 – 10.255.255.255	1 large network	10.0.0.5, 10.1.23.100
Class B	172.16.0.0 – 172.31.255.255	16 medium networks	172.16.10.5, 172.20.200.100
Class C	192.168.0.0 – 192.168.255.255	256 small networks	192.168.0.10, 192.168.100.254

- These ranges were created to save public IP addresses because they were running out.
- Routers block private IP addresses from being routed over the Internet.

Characteristics

- Can be reused by anyone → No need to register them.
- Not routable on the Internet → Must use NAT (Network Address Translation) to connect to the Internet.
- Commonly used in homes, schools, companies to assign IP addresses to many devices without needing many public IPs.

Practical Examples

- **Home Network:**
 - Router: 192.168.1.1
 - Laptop: 192.168.1.10
 - Smartphone: 192.168.1.20
 - Printer: 192.168.1.50
- **Office Network:**
 - Use 10.0.0.0/16 to cover 65,000+ devices.
 - Example addresses: 10.0.1.5 (server), 10.0.2.25 (PC), 10.0.5.100 (printer).
- All devices share one public IP address (via NAT) when accessing the Internet.

4.1. IPv4 ADDRESSES

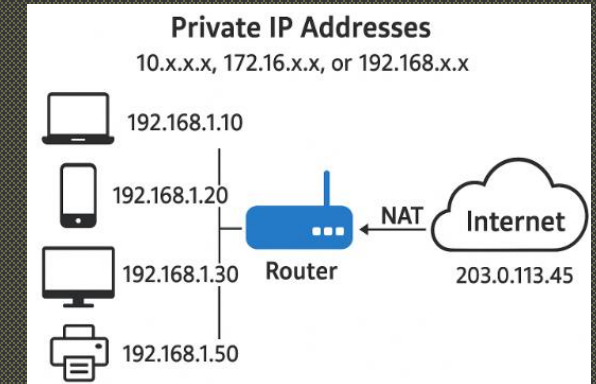
APIPA Addresses

Network Address Translation (NAT)

- Purpose: Translate private IPs into a single public IP for Internet access.
- Example:
 - Your laptop has 192.168.1.50.
 - NAT converts it to 203.0.113.45 (your ISP-assigned public IP) before sending data to the Internet.
 - When a response comes back, NAT sends it back to the correct internal device.

Link-Local / APIPA Addresses

- Range: 169.254.0.0 – 169.254.255.255 (with mask 255.255.0.0)
- Automatic assignment when:
 - Device is set to get an IP via DHCP, but no DHCP server is available.
- Purpose: Allow basic local communication without manual setup.
- Example:
 - Two laptops connected with a cable but no router → both get addresses like 169.254.12.45 and 169.254.88.200 → can still share files locally.



Private IPs Are Useful

- Saves scarce public IP addresses.
- Allows many devices to share one public IP.
- Increases security (outside devices cannot directly connect to private IP devices).
- Flexible - organizations can create as many internal networks as needed.

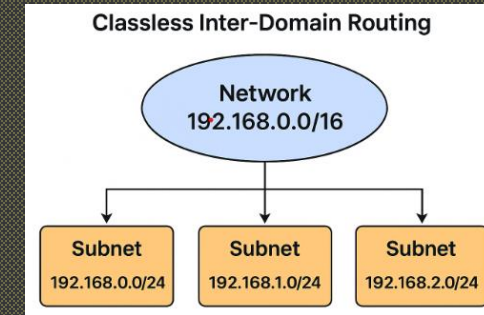
4.1. IPv4 ADDRESSES

Special Service IP Addresses

FROM IP ADDRESS	TO IP ADDRESS	Range Usage
0.0.0.0	0.255.255.255	Reserved
10.0.0.0	10.255.255.255	Class A Private
127.0.0.1	127.255.255.255	Loopback
128.0.0.0	128.0.255.255	Reserved
169.254.0.1	169.254.255.255	Microsoft APIPA
172.16.0.0	172.31.255.255	Class B Private
191.255.0.0	191.255.255.255	Reserved
192.0.0.0	192.0.0.255	Reserved
192.168.0.0	192.168.255.255	Class C Private
223.255.255.0	225.255.255.255	Reserved

4.1. IPv4 ADDRESSES

Classless Inter-Domain Routing

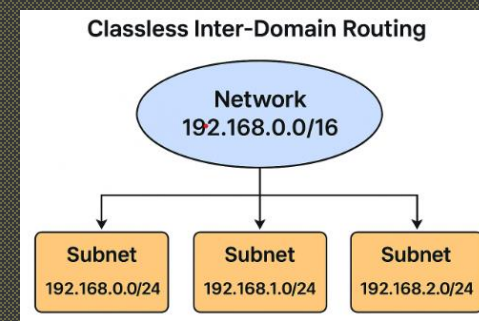


- Classful addressing (A, B, C) was too rigid:
 - It used default subnet masks (e.g., Class B → 255.255.0.0) regardless of network size.
 - Often wasted IP addresses (e.g., giving 65,534 addresses to a network that only needs 100 devices).
- CIDR (Classless Inter-Domain Routing) allows:
 - Custom subnet masks → more efficient use of IP space.
 - Flexible network sizes → small or large networks can get just the addresses they need.
- Features of CIDR
 - Lets you choose your subnet mask instead of using default class masks.
 - Reduces waste of IP addresses.
 - Improves network performance by splitting large networks into smaller subnets.
 - Expresses addresses using CIDR notation:
 - Format: A.B.C.D/n
 - n = number of network bits (1s in subnet mask).
 - Remaining bits = host bits (number of available hosts).
 - Formula for hosts: $2^n - 2$ (subtract network + broadcast address)

IP Address & Mask	CIDR Notation	Network ID Bits	Host Bits	Number of Usable Hosts
172.31.210.10 with 255.255.255.0	172.31.210.10/24	24	8	254 hosts
10.25.106.12 with 255.255.240.0	10.25.106.12/20	20	12	4094 hosts
192.168.1.5 with 255.255.255.192	192.168.1.5/26	26	6	62 hosts

4.1. IPv4 ADDRESSES

Classless Inter-Domain Routing



Advantages of CIDR

- Address Efficiency:
 - Example: If you need 50 IP addresses, you can use /26 (62 usable hosts) instead of a full Class C network (254 hosts).
- Better Network Performance:
 - Large broadcast domains (65,000+ hosts) create excessive broadcast traffic → network slowdown.
 - Subnetting with CIDR limits broadcasts to smaller groups.
- Scalability:
 - ISPs can aggregate multiple small networks into one large prefix (called route aggregation or supernetting), reducing routing table size.

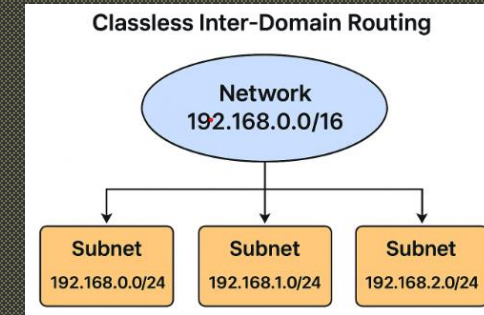
Practical Example

- Suppose a company has 3 departments:
 - HR (20 devices)
 - IT (50 devices)
 - Sales (200 devices)
- Instead of using a single Class B network (wasting thousands of addresses), you can:
 - Assign HR: 192.168.1.0/27 → 30 usable hosts
 - Assign IT: 192.168.1.32/26 → 62 usable hosts
 - Assign Sales: 192.168.1.96/24 → 254 usable hosts
- This way, you use only the addresses you need and keep each department in its own broadcast domain.

IP Address & Mask	CIDR Notation	Network ID Bits	Host Bits	Number of Usable Hosts
172.31.210.10 with 255.255.255.0	172.31.210.10/24	24	8	254 hosts
10.25.106.12 with 255.255.240.0	10.25.106.12/20	20	12	4094 hosts
192.168.1.5 with 255.255.255.192	192.168.1.5/26	26	6	62 hosts

4.1. IPv4 ADDRESSES

Classless Inter-Domain Routing



Broadcast Domains

- Definition: All devices that share the same network ID (and can “hear” each other’s broadcast packets).
- Problem with Large Networks:
 - Every broadcast packet (DHCP, ARP) interrupts all devices in the network.
 - With thousands of devices, CPU and network load skyrockets.
- CIDR + Subnetting Fix:
 - Break one large broadcast domain into many smaller ones, reducing unnecessary traffic.
- Note:
 - More prefix bits → fewer hosts (smaller network).
 - Fewer prefix bits → more hosts (bigger network).

Prefix	Subnet Mask	Usable Hosts
/30	255.255.255.252	2
/29	255.255.255.248	6
/28	255.255.255.240	14
/26	255.255.255.192	62
/24	255.255.255.0	254
/16	255.255.0.0	65,534

4.1. IPv4 ADDRESSES

Subnetting

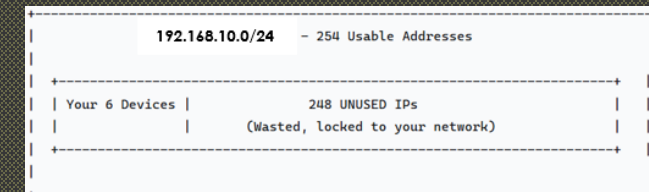
- Subnetting = dividing a large network into smaller networks (subnets).
- Each subnet has its own network ID and a limited number of host addresses.
- Helps make networks more efficient, secure, and manageable.

Reduce broadcast domains

- Example 1: Instead of 65,534 devices in one giant network (172.31.0.0/16), you can create 256 subnets like 172.31.0.0/24, 172.31.1.0/24, ..., 172.31.255.0/24, each with 254 usable IPs.
- Result: Broadcast traffic stays local and doesn't overwhelm the entire network.

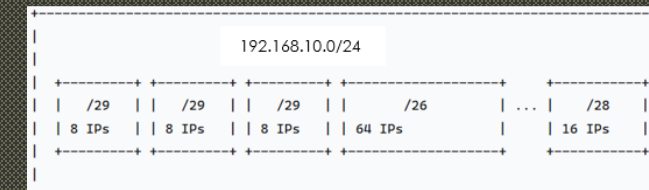
Conserve IP addresses

- Example 2: If you need only 6 public IP addresses, instead of wasting a whole Class C (/24 with 254 IPs), you can use /29 (only 8 IPs: 6 usable, 2 reserved).
- Saves valuable IPv4 address space for others.



Improve security & organization

- Example: Put Payroll Department computers on 192.168.10.0/29 and block access from other subnets using a router ACL (Access Control List).
- Makes it easier to control traffic between departments.



4.1. IPv4 ADDRESSES

Subnetting

Steps to Calculate a Subnet Mask

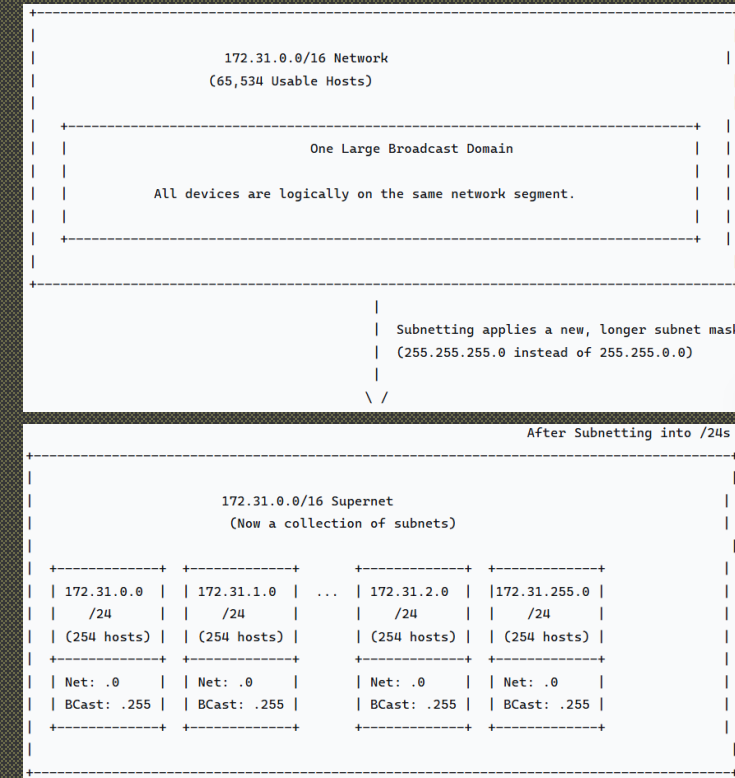
- Decide how many subnets you need
 - Count router interfaces (each needs a unique subnet).
 - Example: You need 20 subnets.
- Find how many bits to “borrow” from host portion
 - Formula: $2^n \geq \text{required subnets}$
 - Example: $2^5 = 32$ (enough for 20 subnets), so borrow 5 bits.
- Recalculate subnet mask
 - Start with default mask. Add borrowed bits.
 - Example: Class B (255.255.0.0) becomes 255.255.252.0 when borrowing 6 bits.
 - In CIDR notation: 172.20.0.0/22.
- Check number of hosts per subnet
 - Formula: $2^n - 2$ (subtract network & broadcast addresses).
 - Example: $2^{10} - 2 = 1022$ hosts per subnet.

Example (Refer to the figures) - Breaking /16 into /24s:

- Original network: 172.31.0.0/16 (65,534 hosts).
- Subnetted into /24s:
 - 172.31.0.0/24 → 172.31.0.1 - 172.31.0.254
 - 172.31.1.0/24 → 172.31.1.1 - 172.31.1.254
 - ...
 - 172.31.255.0/24 → 172.31.255.1 - 172.31.255.254
- Each subnet: 254 usable IPs, small broadcast domain.

Subnetting "borrows" bits from the host portion to create more network prefixes.

- Original (/16): 172.31.0.0 with mask 255.255.0.0
 - Network Bits: 172.31 (16 bits)
 - Host Bits: The last two octets (0.0) - 16 bits for 65,534 hosts.
- New Subnets (/24): 172.31.X.0 with mask 255.255.255.0
 - Network Bits: 172.31.X (24 bits). The third octet (X) is now part of the network address.
 - Host Bits: Only the last octet (8 bits) for 254 hosts per subnet.



4.1. IPv4 ADDRESSES

Subnetting

Finding Subnet Increment

- Increment = place value of the rightmost borrowed bit
 - Example:
 - Borrowed 6 bits in 3rd octet → rightmost bit = 4's place
 - Subnets increase by 4 in 3rd octet:
 - 172.20.0.0, 172.20.4.0, 172.20.8.0, 172.20.12.0, ...
 - The Last Borrowed Bit - Its place value is 4 ($2^2 = 4$).
 - Apply the Increment to Find the Subnets. Original network is 172.20.0.0.
 - The borrowing happens in the 3rd octet, so that's the octet that will change by the increment.
 - 1st Subnet: 172.20.0.0 (The starting point)
 - 2nd Subnet: 172.20.4.0 ($0 + 4 = 4$)
 - 3rd Subnet: 172.20.8.0 ($4 + 4 = 8$)
 - 4th Subnet: 172.20.12.0 ($8 + 4 = 12$)
 - ...and so on: 172.20.16.0, 172.20.20.0, ... all the way up to 172.20.252.0.
- Identify the interesting octet: Where is the subnet mask not 255 or 0? (In this example, the 3rd octet).
 - Find the magic number: $256 - [\text{the value of the interesting octet in your mask}]$.
 - Example: If 6 bits were borrowed, the mask in that octet is 1111100 in binary, which is 252 in decimal.
 - $256 - 252 = 4$. This is the increment.
 - This math shortcut provides the place value of the last borrowed bit without drawing the binary table.
 - In summary: The subnet increment is the heartbeat of subnetting. It tells the consistent jump between subnets.

128	64	32	16	8	4	2	1
2^7	2^6	2^5	2^4	2^3	2^2	2^1	2^0
Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
Borrowed	Borrowed	Borrowed	Borrowed	Borrowed	Borrowed	Host	Host

4.1. IPv4 ADDRESSES

Subnetting

Another Example

- Network: 192.168.100.0/24 (Class C)
- Need for: 4 subnets
- Step 1: Borrow 2 bits ($2^2 = 4$)
- New mask: 255.255.255.192 (/26)
- Each subnet: 62 usable IPs
 - Subnet 1: 192.168.100.0/26 → 192.168.100.1 - 192.168.100.62
 - Subnet 2: 192.168.100.64/26 → 192.168.100.65 - 192.168.100.126
 - Subnet 3: 192.168.100.128/26 → 192.168.100.129 - 192.168.100.190
 - Subnet 4: 192.168.100.192/26 → 192.168.100.193 - 192.168.100.254

Network: 192.168.100.00/24 (Class C)

- You need: 4 subnets
- Step 1: Borrow 2 bits ($2^2 = 4$)
- New mask: 255.255.255.192 (/26)

Subnets

Subnet 1:	Subnet 2:	Subnet 3:	Subnet 4:
192.168.100.026	192.168.100.64/	192.168.100.128/	192.168.100.192/
192.168.100.1 - 192.168.100.62	192.168.100.65 - 192.168.100.126	192.168.100.129 - 192.168.100.190	192.168.100.193 192.168.100.254
62 usable IPs	62 usable IPs	62 usable IPs	62 usable IPs

Summary

- Subnetting creates smaller, efficient, secure networks.
- Reduces broadcast traffic and wasted IPs.
- Use CIDR notation (/n) to quickly express subnet size.
- Formula reminders:
 - Subnets: 2^n (n = borrowed bits)
 - Hosts per subnet: $2^n - 2$ (n = host bits)

Network ID	Hosts Needed	Host Bits	CIDR Notation	Usable IPs
10.19.32.0	900	10	/22	1022
172.25.110.0	505	9	/23	510
192.168.100.32	28	5	/27	30
198.60.123.0	6	3	/29	6 usable

4.1. IPv4 ADDRESSES

Supernetting

- Supernetting = Opposite of subnetting
- Subnetting: Divides a large network into smaller networks
- Supernetting: Combines several smaller networks into one larger network
- Often called route aggregation or route summarization (especially in routing tables)

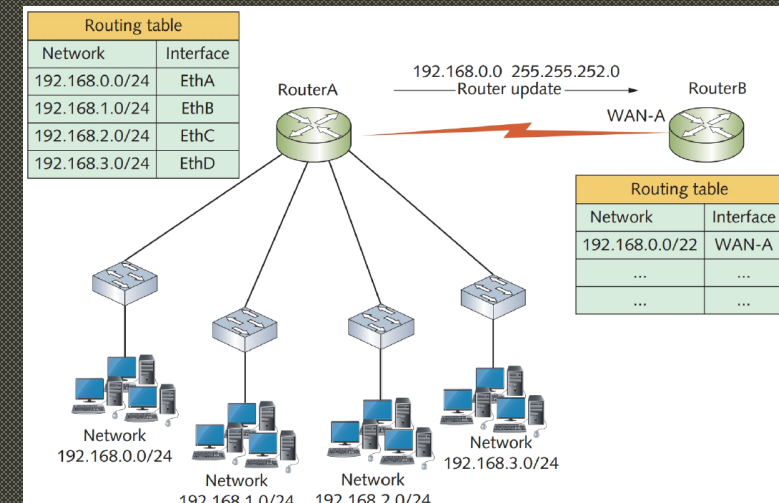
Need for Supernetting:

- To combine multiple small networks into a bigger one
 - Makes IP management simpler
 - Having more hosts than one small network can support
- To make routing tables smaller and faster
 - Instead of having many entries for many networks, you summarize them into one entry
 - Reduces the work routers have to do
 - Improves network efficiency

How Supernetting works:

- Take two or more consecutive network addresses
- Borrow bits from the network portion and add them to the host portion
 - This allows more host addresses per network
- Use a new subnet mask that reflects the larger network

Combined Networks	Original Mask	New Supernet Mask	New CIDR	Total Usable Hosts
192.168.0.0, 192.168.1.0	255.255.255.0	255.255.254.0	/23	510
192.168.0.0 – 192.168.3.0	255.255.255.0	255.255.252.0	/22	1022
10.0.0.0 – 10.0.15.0	255.255.255.0	255.255.240.0	/20	4094



4.1. IPv4 ADDRESSES

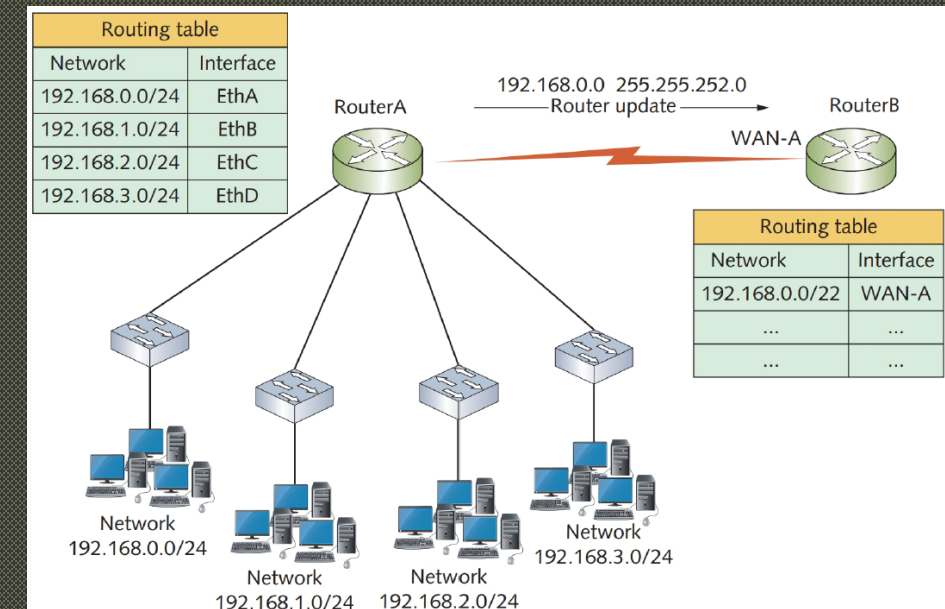
Supernetting

Example 1: Combining Four Class C Networks

- There are 4 Class C networks:
 - 192.168.0.0/24
 - 192.168.1.0/24
 - 192.168.2.0/24
 - 192.168.3.0/24
- We need 900 hosts total but don't want to configure 4 separate subnets

Solution: Supernetting

- Borrow 2 bits from the network portion → combine the networks
- New subnet mask: 255.255.252.0 (or /22)
- Resulting network: 192.168.0.0/22
- Number of host addresses:
 - Host bits = 10 (8 original + 2 borrowed)
 - Formula = $2^{10} - 2 = 1022$ usable hosts
- Enough for all 900 hosts in one single network
- Router B now knows all addresses from 192.168.0.0 → 192.168.3.255 go through Router A



Supernetting always uses consecutive network addresses.

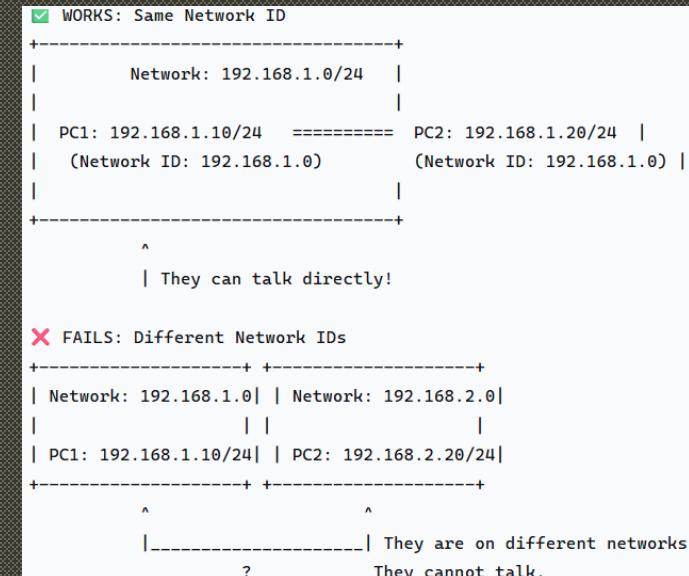
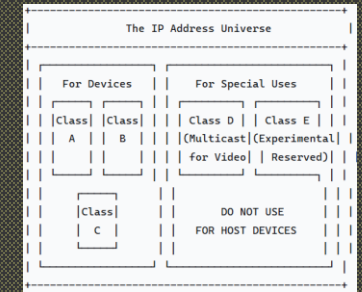
The new mask has fewer 1s than the original → making the network portion smaller.

This means more hosts per network

4.1. IPv4 ADDRESSES

Rules for Assigning IP Addresses

- IP addresses can only be assigned from Class A, B, or C ranges. Class D and Class E addresses are for special use.
- Every IP Address needs a subnet mask. Without a subnet mask, the computer doesn't know if another device is on the same network or not.
 - Examples:
 - 192.168.1.10 with 255.255.255.0 → knows it's part of the 192.168.1.0 network
 - 192.168.1.10 with 255.255.0.0 → thinks it's part of the much larger 192.168.0.0 network
- All hosts on the same network must share the same network ID.
- Computers connected to the same switch (same LAN) must have the same network ID to communicate without a router.
- Example:
 - Works:
 - PC1 → 192.168.1.10/24
 - PC2 → 192.168.1.20/24
 - Same network ID → can talk directly
 - Does NOT Work:
 - PC1 → 192.168.1.10/24
 - PC2 → 192.168.2.20/24
 - Different network IDs → need a router to communicate



Rules for Assigning IP Addresses

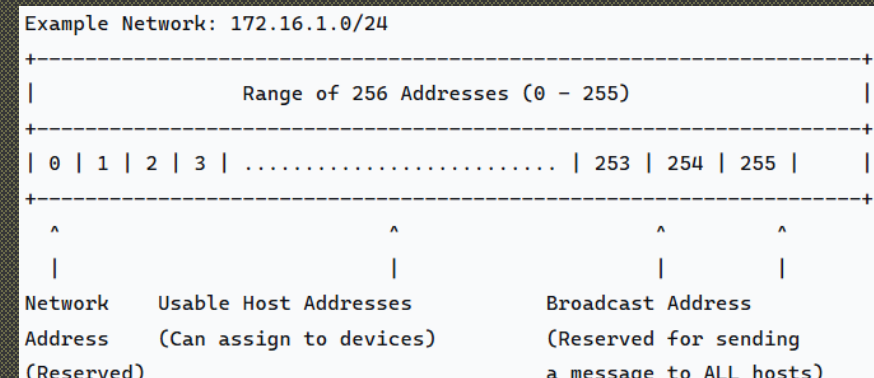
- ```

✔ GOOD: Unique Addresses
Street: Network (192.168.1.0)
| House #10 (PC1) | House #11 (PC2) | House #12 (Printer) |

✖ BAD: IP Conflict
Street: Network (192.168.1.0)
| House #10 (PC1) | House #10 (PC2) | <<< CONFLICT! >>>

"Which one is the real #10?" -> Network confusion and errors

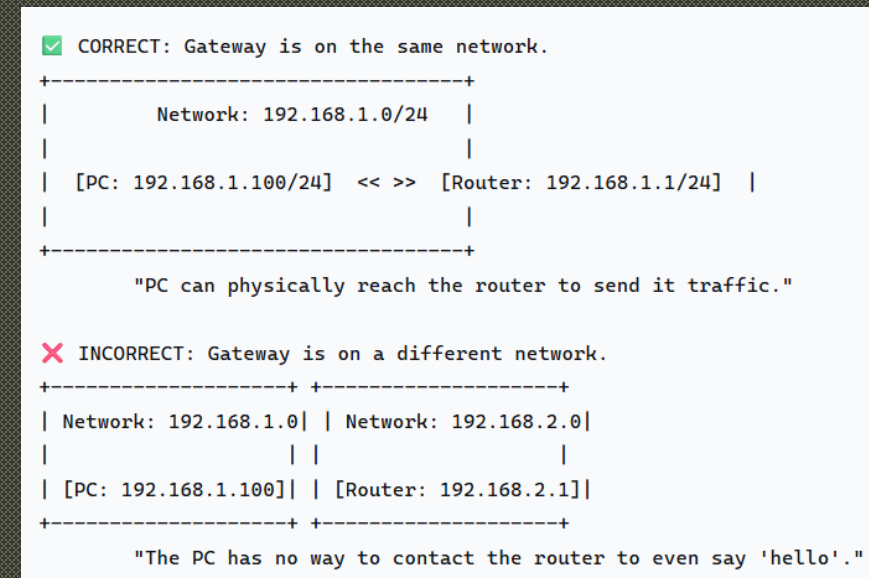
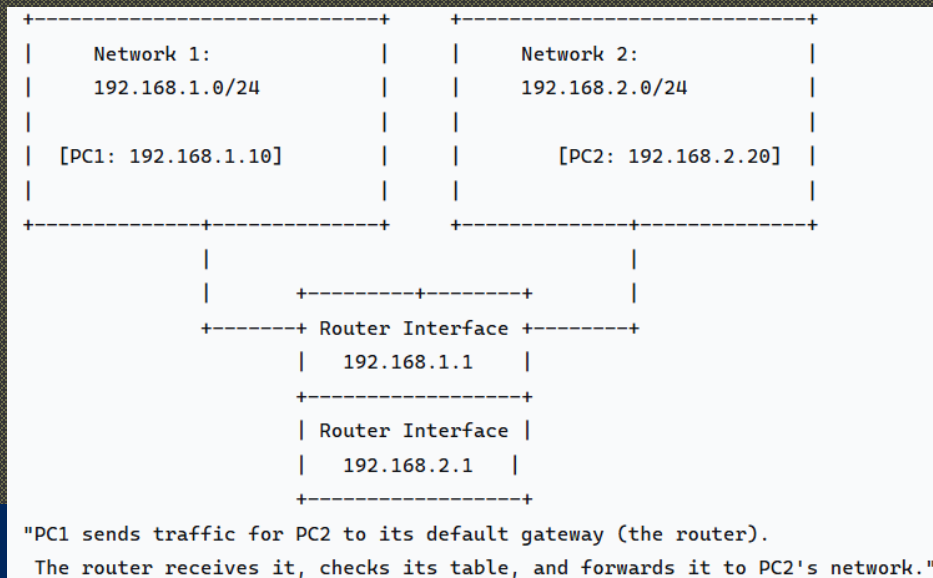
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# 4.1. IPv4 ADDRESSES

## Rules for Assigning IP Addresses

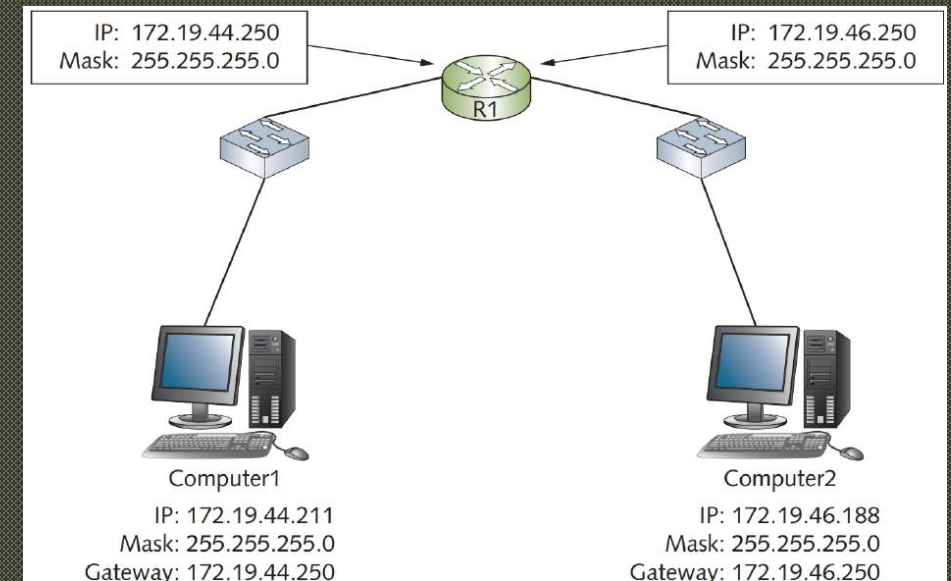
- Computers on different networks need a router to talk; they cannot communicate directly.
  - Example: A router with interfaces in both networks must forward their traffic.
    - PC1 → 192.168.1.10/24
    - PC2 → 192.168.2.10/24
- The computer's IP address and computer's default gateway (router address) must share the same network ID.
  - Example:
    - PC IP: 10.1.0.100/16 → Gateway: 10.1.0.1 - same network ID
    - If the IP address of the Gateway: 10.2.0.1 - different network ID → no communication



# 4.1. IPv4 ADDRESSES

## Configuring the Default Gateway

- The default gateway is usually a router (or computer acting as a router).
- It's the exit point for packets that need to leave the local network.
- Without a default gateway, your computer cannot reach other networks (including the Internet).
- Example:
  - Your PC: 192.168.1.10/24
  - Default gateway: 192.168.1.1 (router)
  - If you try to access Google, packets are sent to 192.168.1.1, which forwards them to the Internet.
- The gateway address must be in the same network as the host.
- Otherwise, the host won't know how to reach it.
  - Example:
    - PC: 192.168.1.10/24, Gateway: 192.168.1.1
      - Same network → Works
    - PC: 192.168.1.10/24, Gateway: 192.168.2.1
      - Different network → PC cannot reach gateway



# 4.1. IPv4 ADDRESSES

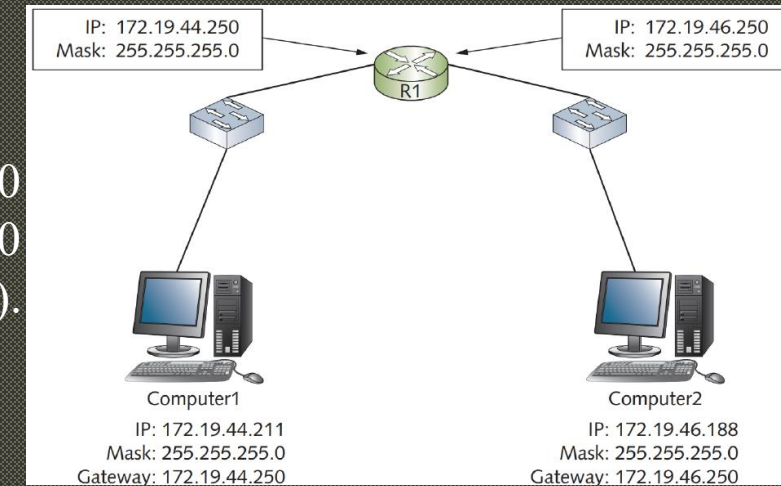
## Configuring the Default Gateway

The computer uses its subnet mask to figure out if the destination is on the same network.

- Computer calculates its own network address (IP AND subnet mask).
- Calculates the network address of the destination IP (IP AND subnet mask).
- If both network addresses match → send packet directly to destination.
- If network addresses are different → send packet to default gateway.
- Example:
  - Computer1: IP = 172.19.44.250, Mask = 255.255.255.0 → Network = 172.19.44.0
  - Computer2: IP = 172.19.46.188, Mask = 255.255.255.0 → Network = 172.19.46.0
  - Networks are different → Computer1 sends packet to its gateway (172.19.44.250).

### Multiple Default Gateways

- You can configure more than one gateway for fault tolerance.
- Windows assigns a metric (priority) based on connection speed and uses the best one first.
- If the main gateway stops responding → Windows automatically switches to backup.
- Later, Windows periodically checks if the main gateway is back (fail-back feature).
- Example:
  - Primary gateway: 192.168.1.1 (fiber connection)
  - Secondary gateway: 192.168.1.2 (backup DSL connection)
  - If main router fails, computer uses backup router automatically.



# 4.1. IPv4 ADDRESSES

## Configuring the Default Gateway

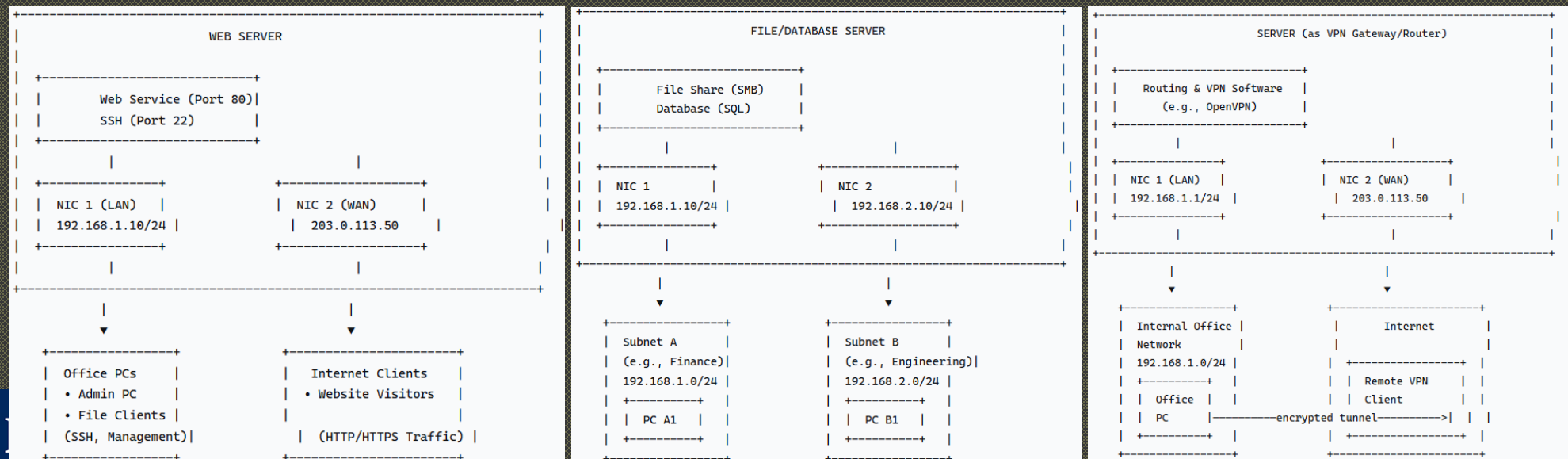
- A multihomed server has two or more network interface cards (NICs), each connected to different networks. Uses:
- Internal + External Access:
  - Internal NIC: 192.168.1.10/24 → used by office PCs
  - External NIC: Public IP → used by Internet clients
- Multiple Subnets:
  - NIC1: 192.168.1.10/24 (Subnet A)
  - NIC2: 192.168.2.10/24 (Subnet B)
  - Allows server to serve both subnets at the same time
- VPN or Router Role:
  - One NIC connected to internal network, another connected to WAN/Internet

A server with 2 or more NICs (Network Interface Cards).

- Each NIC connects to a different network.
- Each NIC needs its own IP address for the network it belongs to.
- Think of it like a person with two phones:
  - One phone for personal calls (internal network).
  - Another phone for work calls (external/Internet).

### Example

- Imagine a small company with a web server:
- NIC1: 192.168.1.10/24 → Internal network (for employees)
- NIC2: Public IP (e.g., 203.0.113.50) → Connected to Internet
- Default gateway is configured on NIC2 (Internet traffic go back out to the Internet).
- No default gateway on NIC1 (so internal traffic stays inside the network).



# 4.1. IPv4 ADDRESSES

## Configuring the Default Gateway

- A default gateway is usually a router (or a computer acting as a router).
- It is the device that your computer sends packets to when the destination is on another network.
- Without a default gateway, computers could only talk to devices on the same local network.

- Example:

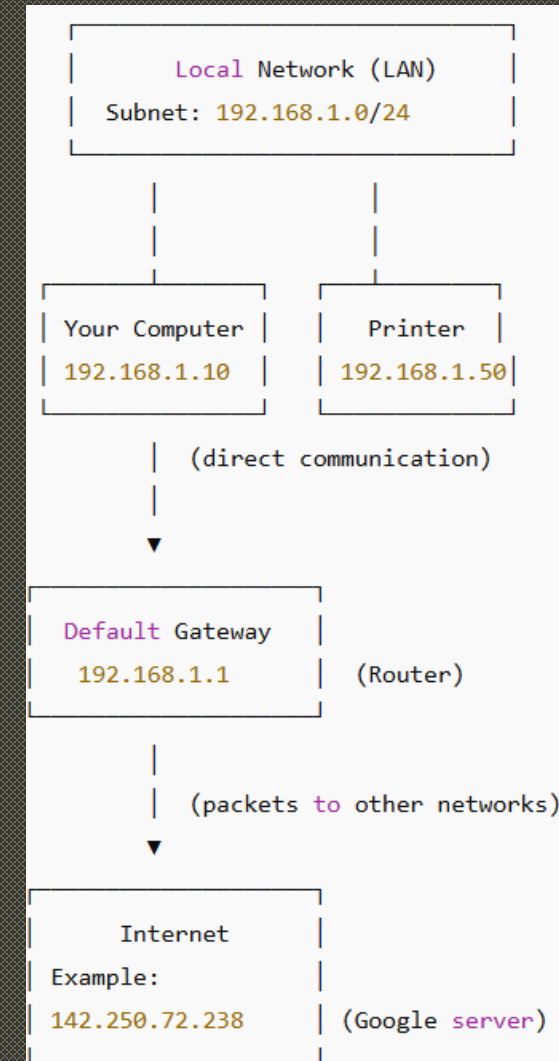
- The computer: 192.168.1.10
- Printer: 192.168.1.50 (same network) → talk directly.
- Website: 142.250.72.238 (Google server, different network) → send to default gateway, which forwards it to the Internet.

The subnet mask helps the computer decide whether to send packets directly or via the default gateway.

- Computer checks if the destination IP in its network range?
  - If yes → send directly.
  - If no → send to default gateway.

- A default gateway is essential for communication outside your local network.
- Computers use the subnet mask to check if the destination is local or remote.
- If remote, the packet is sent to the default gateway.
- Without a properly configured gateway, devices can't reach the Internet or other networks.

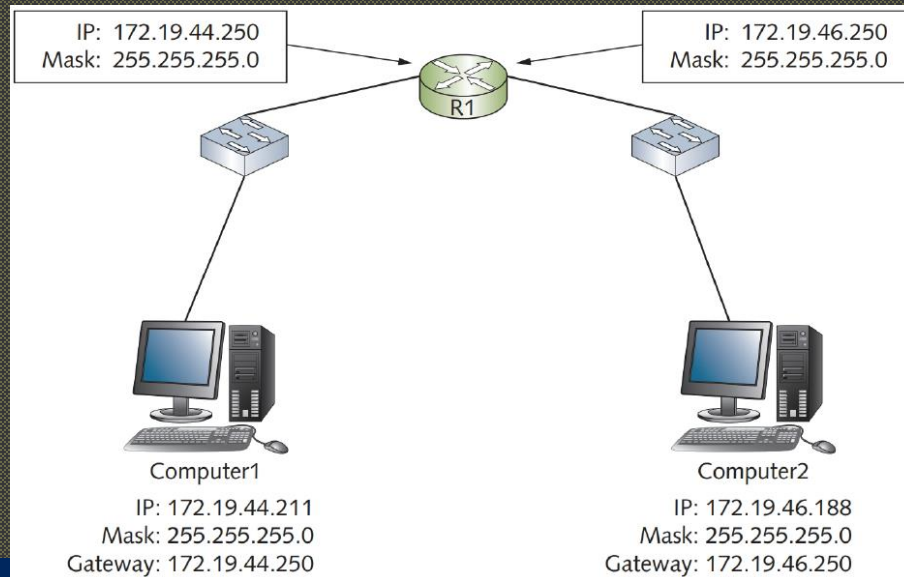
- Computer: 10.0.0.5 / 8 (255.0.0.0)
- Network ID (Computer): 10.0.0.0
  - Destination: 10.20.30.40
    - Network ID (Destination): 10.0.0.0 → Same network → send directly.
  - Destination: 192.168.1.10
    - Network ID (Destination): 192.168.1.0 → Different network → send to default gateway.



# 4.1. IPv4 ADDRESSES

## Configuring the Default Gateway

- Computer1
  - IP: 172.19.44.211
  - Mask: 255.255.255.0 → Network ID: 172.19.44.0
  - Gateway: 172.19.44.250
- Computer2
  - IP: 172.19.46.188
  - Mask: 255.255.255.0 → Network ID: 172.19.46.0
  - Gateway: 172.19.46.250



- Computer1 calculates its own network ID:  
 $172.19.44.211 \text{ AND } 255.255.255.0 \rightarrow 172.19.44.0$
- Computer1 calculates Computer2's network ID:  
 $172.19.46.188 \text{ AND } 255.255.255.0 \rightarrow 172.19.46.0$
- Compare the two network IDs:
  - Computer1 = 172.19.44.0
  - Computer2 = 172.19.46.0
  - Conclusion: Different networks.
- Decision:
  - Computer1 sends the packet to its default gateway (172.19.44.250).
  - The gateway forwards it to 172.19.46.250, which delivers it to Computer2.
- The default gateway must be in the same network as the host.
- This is determined using the subnet mask.
- If not, the computer cannot reach its own gateway.
  - Example:
    - Computer: 192.168.1.25 /24 (255.255.255.0)
    - Valid gateway must be between 192.168.1.1 and 192.168.1.254.
    - If 192.168.2.1 is configured as the gateway → it will not work, because it is a different network.

# 4.1. IPv4 ADDRESSES

## Using Multiple Default Gateways

- A default gateway = the router a computer uses to send packets outside its network.
- Normally, it is one default gateway.
- But Windows allows multiple configuration of multiple default gateways for backup/fault tolerance.
  - Primary Gateway → your computer uses it first to send traffic.
  - Metric Value → Windows assigns a number to each gateway.
    - Lower metric = preferred (faster/better route).
  - If primary gateway fails (router down, cable unplugged, etc.), Windows:
    - Switches automatically to the secondary gateway.
  - Fail-back feature:
    - Windows keeps checking the primary gateway.
    - If it comes back online, Windows switches back to it.

### Example: Home Network

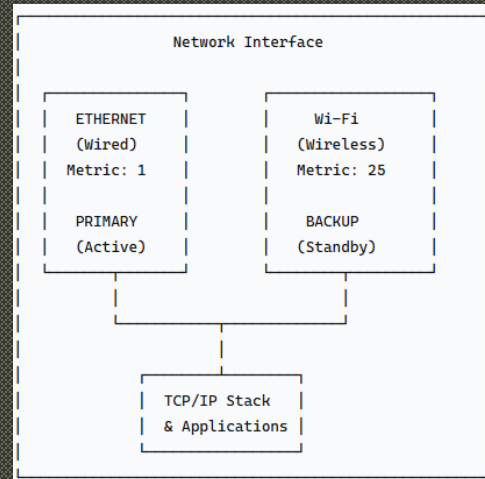
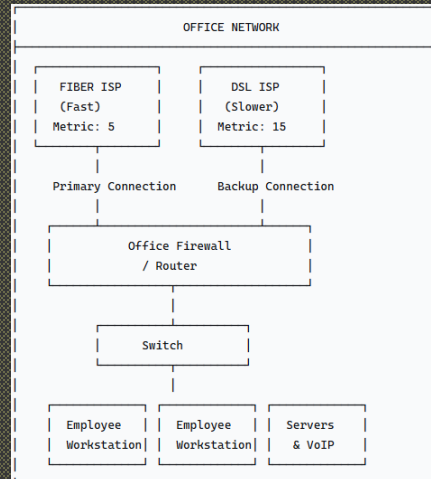
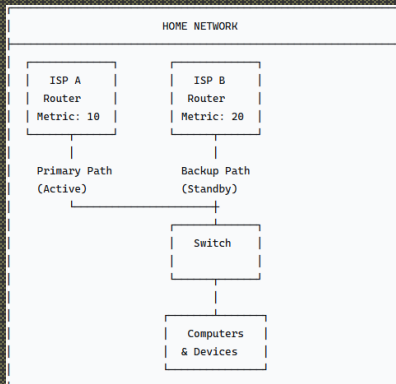
- Gateway 1: Router from ISP A → Metric 10.
- Gateway 2: Backup router from ISP B → Metric 20.
- If ISP A router fails, Windows uses ISP B's router.

### Example: Office Network

- Gateway 1: Fiber connection (fast) → Metric 5.
- Gateway 2: DSL connection (slower backup) → Metric 15.
- Employees still have Internet even if the fiber goes down.

### Example: Laptop with Wired + Wi-Fi

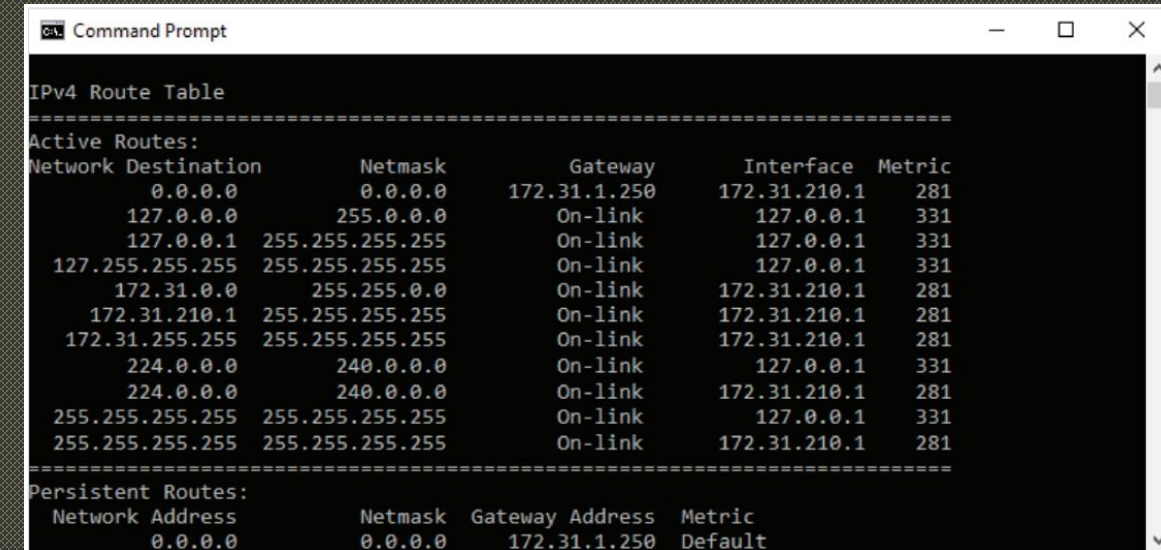
- Gateway 1: Ethernet (wired, faster) → Metric 1.
- Gateway 2: Wi-Fi (wireless, slower) → Metric 25.
- Windows prefers Ethernet. If you unplug the cable, it automatically switches to Wi-Fi.



# 4.1. IPv4 ADDRESSES

## route Command

- Windows computers keep a routing table (a list of rules for sending packets to the right destination).
- The route.exe command lets you view and change this routing table.
- Command:
  - route print → shows the current routing table.
  - route add → adds a new route.
  - route delete → removes a route.
- Network Destination → The destination network or IP address.
  - Example: 172.31.0.0 means packets going to that network.
- Netmask → Defines the network portion of the address.
  - Example: 255.255.0.0 means /16.
- Gateway → The router (next hop) to send packets through.
  - Example: 172.31.1.250 is the default gateway.
- Interface → The local NIC (Network Interface Card) used to send the packet.
- Metric → Priority of the route (lower = preferred).
  - Example: If two routes can reach the same place, the one with the lowest metric is chosen.



Command Prompt

IPv4 Route Table

=====

Active Routes:

| Network Destination | Netmask         | Gateway      | Interface    | Metric |
|---------------------|-----------------|--------------|--------------|--------|
| 0.0.0.0             | 0.0.0.0         | 172.31.1.250 | 172.31.210.1 | 281    |
| 127.0.0.0           | 255.0.0.0       | On-link      | 127.0.0.1    | 331    |
| 127.0.0.1           | 255.255.255.255 | On-link      | 127.0.0.1    | 331    |
| 127.255.255.255     | 255.255.255.255 | On-link      | 127.0.0.1    | 331    |
| 172.31.0.0          | 255.255.0.0     | On-link      | 172.31.210.1 | 281    |
| 172.31.210.1        | 255.255.255.255 | On-link      | 172.31.210.1 | 281    |
| 172.31.255.255      | 255.255.255.255 | On-link      | 172.31.210.1 | 281    |
| 224.0.0.0           | 240.0.0.0       | On-link      | 127.0.0.1    | 331    |
| 224.0.0.0           | 240.0.0.0       | On-link      | 172.31.210.1 | 281    |
| 255.255.255.255     | 255.255.255.255 | On-link      | 127.0.0.1    | 331    |
| 255.255.255.255     | 255.255.255.255 | On-link      | 172.31.210.1 | 281    |

=====

Persistent Routes:

| Network Address | Netmask | Gateway Address | Metric  |
|-----------------|---------|-----------------|---------|
| 0.0.0.0         | 0.0.0.0 | 172.31.1.250    | Default |

0.0.0.0 destination → Default route/gateway - It indicates that the default route covers all possible IP addresses.

Used if no other entry matches. Example: If destination is unknown, packets go to the default gateway.

127.x.x.x → Loopback (refers to "this computer").

224.x.x.x → Multicast addresses (send to multiple systems).

255.x.x.x → Broadcast addresses (send to all devices in the network).

On-link → Means the network is directly connected to the computer (no gateway needed).

# 4.1. IPv4 ADDRESSES

## route Command

- 0.0.0.0 (Default Route)
- Netmask: 0.0.0.0
- Gateway: 172.31.1.250
- Interface: 172.31.210.1
- Metric: 281
- This is the default route, often referred to as the default gateway. It applies to any destination IP address that doesn't match a more specific route in the table. Packets with unmatched destinations are sent to the gateway at 172.31.1.250 via the interface 172.31.210.1. The high metric (281) suggests this is a standard default route, possibly adjusted by the system.
- 127.0.0.0
- Netmask: 255.0.0.0
- Gateway: On-link
- Interface: 127.0.0.1
- Metric: 331
- This is the loopback network, reserved for local communication on the host. The "On-link" gateway indicates that traffic to this network (e.g., 127.0.0.1, the loopback address) is handled locally without involving a router. The metric (331) is higher, reflecting its local nature.
- 127.0.0.1
- Netmask: 255.255.255.255
- Gateway: On-link
- Interface: 127.0.0.1
- Metric: 331
- This is a specific route for the loopback address (127.0.0.1), used for testing and local communication. Like the broader 127.0.0.0 range, it is handled locally.
- 172.31.0.0
- Netmask: 255.255.0.0
- Gateway: On-link
- Interface: 172.31.210.1
- Metric: 281
- This route covers the entire 172.31.0.0/16 network, which includes the local subnet. The "On-link" status indicates that devices in this range are directly accessible via the interface 172.31.210.1, likely the primary network interface.
- 224.0.0.0
- Netmask: 240.0.0.0
- Gateway: On-link
- Interface: 172.31.210.1
- Metric: 281
- This range (224.0.0.0/4) is reserved for multicast addresses. The "On-link" status means multicast traffic is sent locally on the 172.31.210.1 interface, not routed.
- 255.255.255.255
- Netmask: 255.255.255.255
- Gateway: On-link
- Interface: 172.31.210.1
- Metric: 281
- This is the broadcast address for the local network. Packets sent to 255.255.255.255 are broadcast to all devices on the local 172.31.210.0/24 subnet.

```
Command Prompt

IPv4 Route Table
=====
Active Routes:
Network Destination Netmask Gateway Interface Metric

0.0.0.0 0.0.0.0 172.31.1.250 172.31.210.1 281
127.0.0.0 255.0.0.0 On-link 127.0.0.1 331
127.0.0.1 255.255.255.255 On-link 127.0.0.1 331
127.255.255.255 255.255.255.255 On-link 127.0.0.1 331
172.31.0.0 255.255.0.0 On-link 172.31.210.1 281
172.31.210.1 255.255.255.255 On-link 172.31.210.1 281
172.31.255.255 255.255.255.255 On-link 172.31.210.1 281
224.0.0.0 240.0.0.0 On-link 127.0.0.1 331
224.0.0.0 240.0.0.0 On-link 172.31.210.1 281
255.255.255.255 255.255.255.255 On-link 127.0.0.1 331
255.255.255.255 255.255.255.255 On-link 172.31.210.1 281

Persistent Routes:
Network Address Netmask Gateway Address Metric

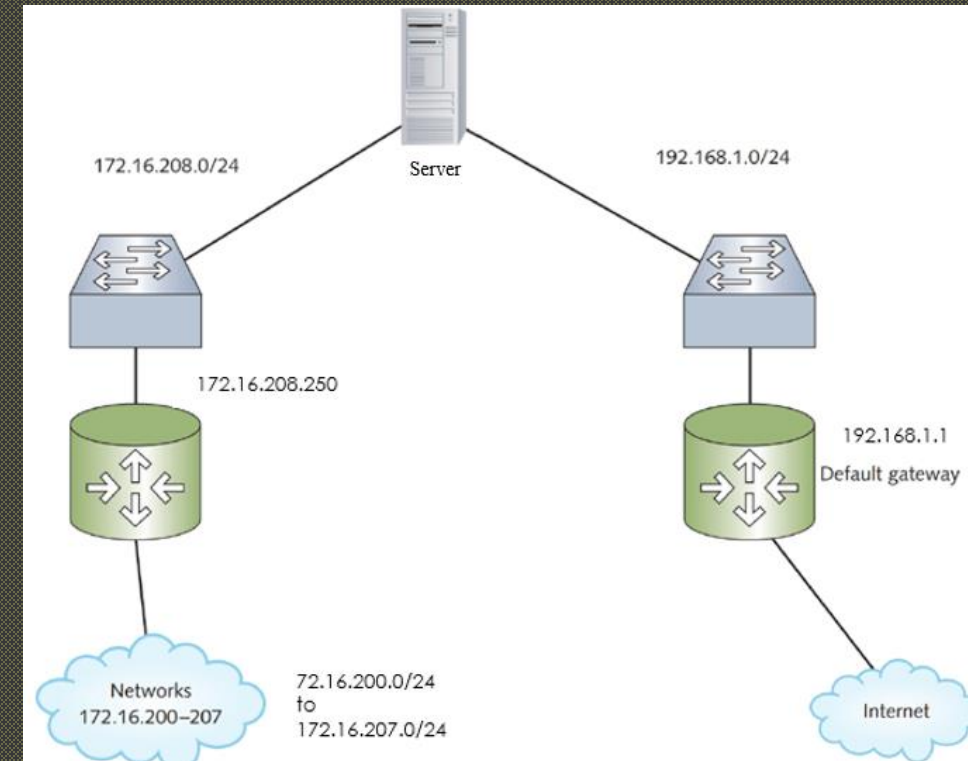
0.0.0.0 0.0.0.0 172.31.1.250 Default
```

# 4.1. IPv4 ADDRESSES

## route Command

A multihomed server (a server with more than one network interface) is connected to two separate networks. The server's routing table is incomplete, causing it to send responses for the internal 172.16.200.0/21 networks out the wrong interface, towards the Internet, where they are lost. The solution is to add specific routes to the server's routing table to direct this internal traffic correctly.

- Interface 1: Connected to 192.168.1.0/24
  - Purpose: This network provides the connection to the Internet.
  - Default Gateway: The server's default gateway (e.g., 192.168.1.1) is correctly configured on this interface. The default gateway is the "gate of last resort" for any traffic destined for a network not found in the routing table (like the Internet).
- Interface 2: Connected to 172.16.208.0/24
  - Purpose: This is part of a larger internal network. A router on this network (with IP 172.16.208.250) provides pathways to other internal subnets: 172.16.200.0/24 to 172.16.207.0/24.
- The Problem:
  - By default, the server's routing table only knows about:
    - Its directly connected networks (192.168.1.0/24 and 172.16.208.0/24).
    - The default route (0.0.0.0/0) pointing to the Internet gateway on 192.168.1.0/24.
  - When the server receives a packet from a computer on, for example, the 172.16.200.10 network, it needs to send a reply to 172.16.200.10. The server checks its routing table:
    - Is 172.16.200.10 on my local 172.16.208.0/24 network? No.
    - Do I have a specific route for the 172.16.200.0/24 network? No.
    - Therefore, the server follows the default route and sends the reply packet out the 192.168.1.0/24 interface towards the Internet. The internal network 172.16.200.0/24 is not reachable from the Internet, so the packet is lost, and communication fails.

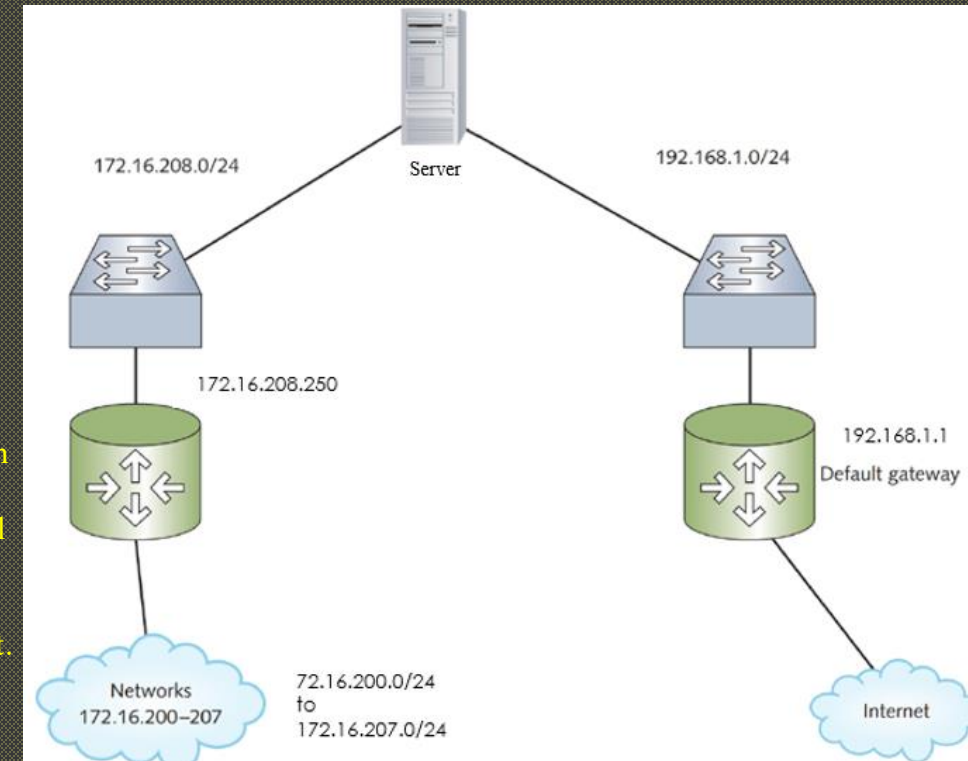


# 4.1. IPv4 ADDRESSES

## route Command

A multihomed server (a server with more than one network interface) is connected to two separate networks. The server's routing table is incomplete, causing it to send responses for the internal 172.16.200.0/21 networks out the wrong interface, towards the Internet, where they are lost. The solution is to add specific routes to the server's routing table to direct this internal traffic correctly.

- To fix this, we need to teach the server how to reach the other internal 172.16.x.0 networks.
- We do this by adding static routes to the routing table using the route add command.
- Command Syntax: `route add <destination_network> mask <subnet_mask> <gateway_ip>`
- Solution: Adding Individual Routes (172.16.200.0)
  - `route add 172.16.200.0 mask 255.255.255.0 172.16.208.250`
    - `route add`: The command to add a new route.
    - 172.16.200.0: The destination network we want the server to know how to reach.
    - mask 255.255.255.0: The subnet mask of the destination network (a /24).
    - 172.16.208.250: The gateway or "next-hop" router - the IP address of the router on the server's own local network (172.16.208.0/24) that can forward packets towards the destination network.
- After running this command, when the server needs to send a packet to any IP in the 172.16.200.0/24 range, it will find this specific route in its table and correctly send the packet out the 172.16.208.0/24 interface to the router at 172.16.208.250.
- We need to add eight separate commands to cover all networks from 172.16.200.0 to 172.16.207.0. Not inefficient.

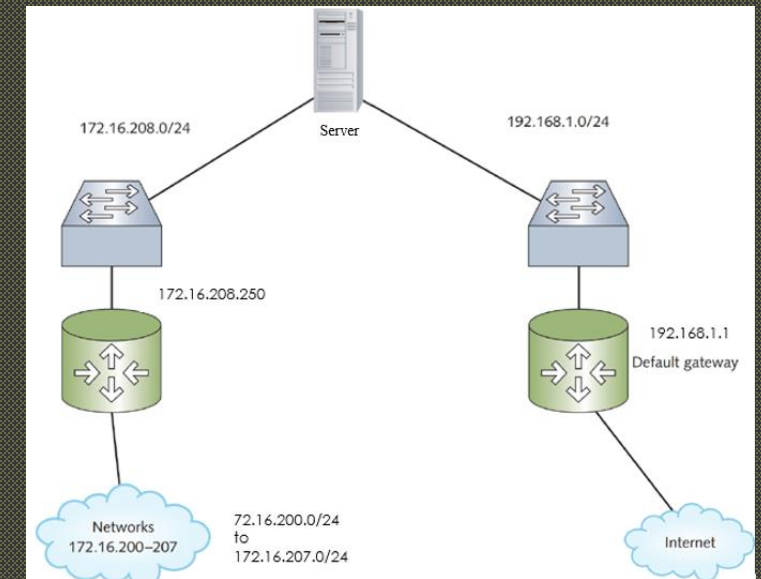


# 4.1. IPv4 ADDRESSES

## route Command

A multihomed server (a server with more than one network interface) is connected to two separate networks. The server's routing table is incomplete, causing it to send responses for the internal 172.16.200.0/21 networks out the wrong interface, towards the Internet, where they are lost. The solution is to add specific routes to the server's routing table to direct this internal traffic correctly.

- A more efficient method is to summarize all eight networks into one route entry using a broader subnet mask.
  - route add 172.16.200.0 mask 255.255.248.0 172.16.208.250
    - Destination Network: 172.16.200.0
    - New Subnet Mask: 255.255.248.0 (which is a /21 in CIDR notation).
      - Look at the third octet of the network addresses in binary:
      - 200 in binary: 1100 1000
      - 201 in binary: 1100 1001
      - ...up to...
      - 207 in binary: 1100 1111
    - A mask of 255.255.255.0 (/24) looks at all 8 bits of the third octet. Each network is different.
    - A mask of 255.255.248.0 (/21) only looks at the first 5 bits of the third octet (because 248 in binary is 1111 1000). The last 3 bits are ignored for the network ID.
    - For all networks from 200 to 207, the first 5 bits of the third octet are 11001 (which is 200).
    - Therefore, the single network address 172.16.200.0 with a mask of 255.255.248.0 encompasses all IP addresses from 172.16.200.1 to 172.16.207.254



| Destination IP           | Before Adding Route (Uses Default Gateway) | After Adding Consolidated Route (Uses Correct Path)    |
|--------------------------|--------------------------------------------|--------------------------------------------------------|
| 8.8.8.8 (Internet)       | Out 192.168.1.x interface (Correct)        | Out 192.168.1.x interface (Correct)                    |
| 172.16.208.5 (Local)     | Out 172.16.208.x interface (Correct)       | Out 172.16.208.x interface (Correct)                   |
| 172.16.203.15 (Internal) | Out 192.168.1.x interface (Fails)          | Out 172.16.208.x interface to 172.16.208.250 (Correct) |

# 4.1. IPv4 ADDRESSES

## netsh Command

- Netsh (net-shell) Command is a Windows command-line tool for network configuration and troubleshooting.
- Allows the user to configure settings like IP addresses, DNS servers, firewalls, and interfaces.
  - Useful for administrators who want to quickly set or change network settings.

### Set a Static IP Address

- Command: `netsh interface ipv4 set address "Ethernet0" static 10.1.1.1 255.255.0.0`
- Meaning:
  - Interface = Ethernet0
  - IP address = 10.1.1.1
  - Subnet mask = 255.255.0.0
- The PC now always uses 10.1.1.1 instead of getting an IP from DHCP.

### Set a Static IP Address + Default Gateway

- Command: `netsh interface ipv4 set address "Ethernet0" static 10.1.1.1 255.255.0.0 10.1.1.250`
- Meaning:
  - Interface = Ethernet0
  - IP address = 10.1.1.1
  - Subnet mask = 255.255.0.0
  - Default gateway = 10.1.1.250
- Traffic that doesn't belong to the local network will go through 10.1.1.250 (the router).

### Set a DNS Server

- Command: `netsh interface ipv4 set dns "Ethernet0" static 10.1.1.100 primary`
- Meaning:
  - Interface = Ethernet0
  - DNS server = 10.1.1.100
  - The IP address 10.1.1.100 is designated as the primary DNS server, meaning it will be the first server the system queries to resolve domain names (e.g., turning "google.com" into an IP address).
- Example: Instead of using Google DNS (8.8.8.8), your PC will now ask 10.1.1.100 for domain lookups.

# 4.1. IPv4 ADDRESSES

## ipconfig Command

- Ipconfig is a Windows command-line tool that shows and manages IP address settings.
  - Often the first tool to use for network troubleshooting.
  - Fast troubleshooting for IP and DNS issues.
  - Helps diagnose why a computer cannot connect to the internet.

### Basic IP Info (no options)

- Command: ipconfig
- Shows:
  - IP address
  - Subnet mask
  - Default gateway

```
IPv4 Address. : 192.168.1.25
Subnet Mask : 255.255.255.0
Default Gateway : 192.168.1.1
```

### Detailed Info (/all)

- Command: ipconfig /all
- Shows extra details:
  - Computer name & domain
  - Network adapter description
  - Physical (MAC) address
  - DHCP status (enabled/disabled)
  - DNS servers
- Example: Useful for checking if your PC is using DHCP or a static IP.

### Clear DNS Cache (/flushdns)

- Command: ipconfig /flushdns
- Action: Deletes old/incorrect DNS info from memory.
- Example: If a website recently changed servers, flushing ensures your PC gets the new IP instead of the old one.

### Re-register with DNS (/registerdns)

- Command: ipconfig /registerdns
- Action: Requests new DHCP lease + updates DNS records.
- Example: Useful when your PC name or IP changes and needs to update DNS servers.

### Release IP (/release)

- Command: ipconfig /release
  - Action: Gives up the current IP (if using DHCP).
  - PC temporarily has 0.0.0.0 as its IP.
- Example: Used before reconnecting or troubleshooting DHCP issues.

### Renew IP (/renew)

- Command: ipconfig /renew
  - Action: Request a new IP address from DHCP.
- Example: If your Wi-Fi disconnects and reconnects but doesn't get an IP, this can fix it.

# 4.1. IPv4 ADDRESSES

## ping Command

- Ping is a tool to test network connectivity between your computer and another device (by IP or hostname).
- Works by sending an ICMP Echo Request → if the other device is reachable, it replies with an ICMP Echo Reply.
- Useful for checking if a computer, server, or router is online and reachable.
- To measure latency (how fast packets travel).
- To troubleshoot network drops or delays.
- To check if DNS resolution works (try ping google.com)

### Ping continuously (-t)

- Command:
- ping 8.8.8.8 -t
- Keeps sending pings until you press Ctrl+C.
- Example: Used to monitor if a connection drops over time.

### Example of a Ping Reply

```
Reply from 192.168.100.201: bytes=32 time<1ms TTL=128
```

- 192.168.100.201 → IP address of the computer that replied.
- bytes=32 → size of the ICMP message (default = 32 bytes).
- time<1ms → how long it took to get a reply (network latency).
- TTL=128 → Time To Live; how many routers the packet can pass through before being dropped.

### Set TTL (-i TTL)

- Command:
- ping 8.8.8.8 -i 5
- TTL = 5 means the packet can only pass through 5 routers before being discarded.
- Example: Helps test how far packets can travel in the network.

### Set number of pings (-n count)

- Command:
- ping 8.8.8.8 -n 10
- Sends 10 packets instead of the default 4.
- Example: Helps test network stability over multiple tries.

### Change packet size (-l size)

- Command:
- ping 8.8.8.8 -l 1000
- Sends a packet of 1000 bytes instead of 32.
- Example: Used to test if large packets (MTU issues) are causing problems.

MTU (Maximum Transmission Unit) - The largest size (in bytes) of a packet that can be sent over a network link without being broken into smaller pieces (fragmented).

- Default MTU on Ethernet = 1500 bytes (most networks use this).
- If a packet is larger than the MTU, it must be fragmented (split up), which slows down communication.

# 4.1. IPv4 ADDRESSES

## arp Command

- ARP (Address Resolution Protocol) maps IP addresses → MAC addresses.
- Needed because:
  - Computers use IP addresses to identify each other.
  - Actual data transfer on a local network uses MAC addresses (hardware addresses).
- Example: To send data to 192.168.1.20, your computer first needs that device's MAC address.

### ARP Process

- Your PC asks: *"Who has IP 192.168.1.20?"* (ARP request).
- That device replies: *"I'm 192.168.1.20, my MAC is 00-1A-2B-3C-4D-5E."*
- Your PC saves that in its ARP cache for faster communication.
- Cache entries expire after a few minutes to avoid using outdated info.

#### View ARP Cache (-a or -g)

- Command:
  - `arp -a`
- Shows IP-to-MAC address mappings currently stored.
- Example output:

| Internet Address | Physical Address  | Type    |
|------------------|-------------------|---------|
| 192.168.1.1      | 00-14-22-01-23-45 | dynamic |
| 192.168.1.20     | 00-1a-2b-3c-4d-5e | dynamic |

#### Delete ARP Cache (-d)

- Command:
  - `arp -d`
- Deletes all entries in the ARP cache.
- Or delete one entry:
  - `arp -d 192.168.1.20`
- Example: Useful if a network card was replaced and has a new MAC address, but your PC still has the old one cached.

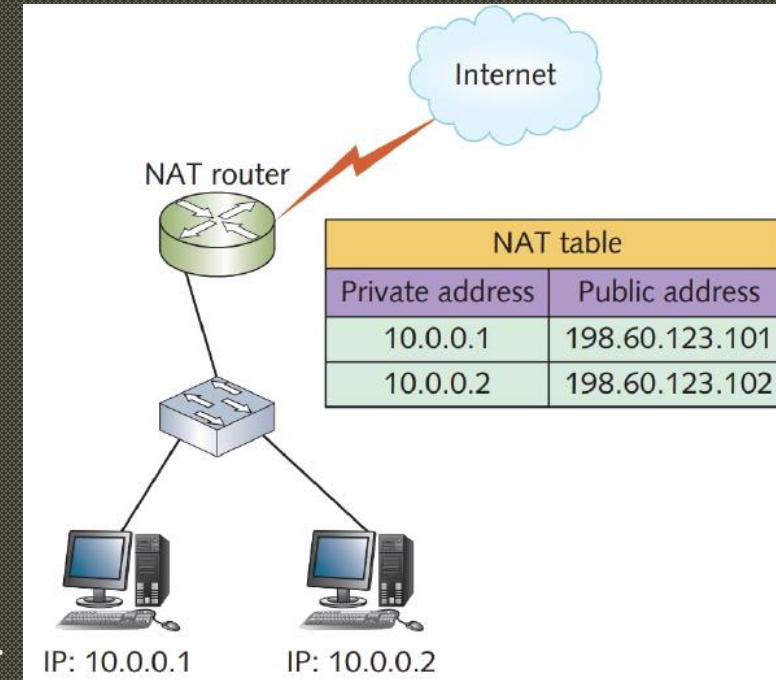
#### 3.Add Permanent Entry (-s)

- Command:
  - `arp -s 192.168.1.50 00-25-96-12-34-56`
- Adds a permanent mapping of IP → MAC.
- Example: If you frequently connect to a server with a fixed MAC, this avoids repeated ARP lookups.
- Note: If the server's NIC is replaced, the MAC will change and this entry becomes invalid.

# 4.1. IPv4 ADDRESSES

## Network Address Translation (NAT)

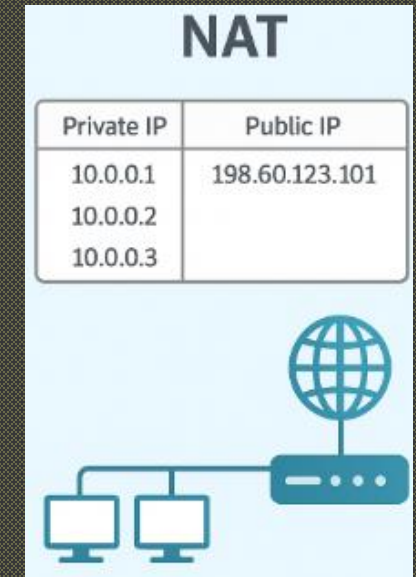
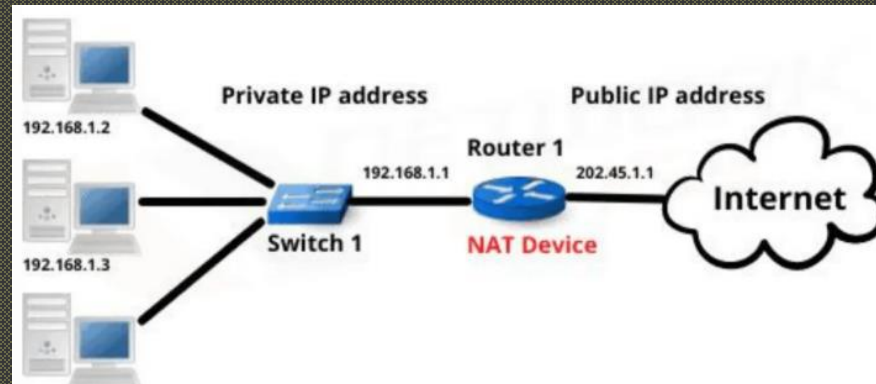
- IPv4 addresses are limited → not enough public IPs for every device in the world.
- Subnetting helps by using addresses more efficiently, but it doesn't create new addresses.
- NAT allows organizations to use private IP addresses inside their network.
- Private IP ranges (not routable on the Internet):
  - Class A: 10.0.0.0 – 10.255.255.255
  - Class B: 172.16.0.0 – 172.31.255.255
  - Class C: 192.168.0.0 – 192.168.255.255
- NAT translates private IPs into public IPs when devices access the Internet.
- Usually handled by a router or firewall at the network's edge.
- A workstation with IP = 10.0.0.1 sends a packet to the Internet.
- The NAT router changes the source IP from 10.0.0.1 → 198.60.123.101 (a public IP).
- When the reply comes back to 198.60.123.101, the router translates it back to 10.0.0.1.
- Result:
  - Internal computers can use private addresses (no conflicts).
  - Only one public IP is required for outside communication.



# 4.1. IPv4 ADDRESSES

## Network Address Translation (NAT)

- Private IP Address Usage:
  - Companies can use private IP addresses within their internal network.
- Public IP Address Requirement:
  - A public IP address is needed only when a workstation accesses the Internet (Humber for the web server).
- NAT (Network Address Translation) Benefits:
  - Reduces the number of public IP addresses required. Not all company's PCs need to go online.
  - One public IP address is needed only when a computer connects to the Internet.
- NAT Drawback:
  - Requires one public IP address for each private IP address in use.
- Typical Use Cases for Public IP Addresses:
  - Primarily needed for devices like web servers or others that must be accessible via the Internet.



# 4.1. IPv4 ADDRESSES

## Port Address Translation (PAT)

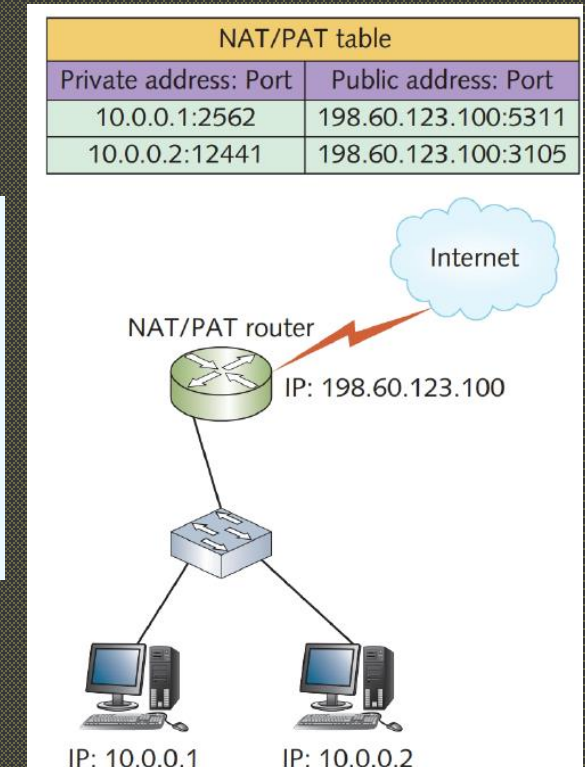
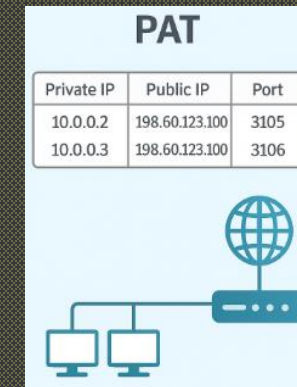
- PAT is extension of NAT that allows many private devices to share a single public IP.
- Works by using port numbers in addition to IP addresses to keep sessions separate.
- Source ports are unique per connection (range: 1024 - 65535).
- Router has public IP = 198.60.123.100.
- Two internal workstations connect to the Internet:
  - 10.0.0.2 → mapped as 198.60.123.100:3105
  - 10.0.0.3 → mapped as 198.60.123.100:3106
- Both share the same public IP, but different port numbers distinguish them.
- When replies come back:
  - Packet to 198.60.123.100:3105 → router knows it belongs to 10.0.0.2.
  - Packet to 198.60.123.100:3106 → router knows it belongs to 10.0.0.3.

Result:

- Hundreds of computers can access the Internet with just one public IP.
- This is what most home and small business routers do by default.

Your home network:

- Phone = 192.168.1.10
- Laptop = 192.168.1.11
- Smart TV = 192.168.1.12
- To the Internet, all appear as one public IP (the router's IP). PAT uses different port numbers to keep the sessions separate.



## 4.2. IPv6 ADDRESSES

- IPv4 has been the main Internet protocol for decades but is running out of address space and requires workarounds for security and quality of service.
  - IPv6 was developed as its replacement, offering larger address space, built-in hierarchy, and structured fields.
  - Configuring IPv6 is different from IPv4.
  - The transition from IPv4 to IPv6 is gradual, so coexistence methods are used to allow communication between both.
- 
- IPv4 has been used for decades
    - Example: Most devices you use today (laptops, phones, routers) are still running IPv4.
  - Problems with IPv4:
    - Address space is running out → Only about 4.3 billion addresses exist.
      - Example: Every device needs a unique IP. With billions of phones, laptops, and IoT devices, IPv4 cannot cover everyone.
    - Requires workarounds for security and quality of service.
      - Example: NAT (Network Address Translation) was added later because IPv4 didn't have built-in security.
  - IPv6 (replacement for IPv4) is created to solve IPv4's problems
    - Provides a much larger address space (about 340 undecillion addresses).
    - Has a hierarchical structure → makes routing more efficient.
    - Built-in features for security and quality of service.
  - Address format looks different
    - IPv4: 192.168.1.1
    - IPv6: 2001:0db8:85a3:0000:0000:8a2e:0370:7334
  - Configuration is different
    - Example: With IPv6, some addresses can configure themselves automatically.
- 
- Transition from IPv4 → IPv6
  - Too many existing IPv4 systems are in use.
  - Coexistence is possible → Special methods allow communication between IPv4 and IPv6.
    - Example: Dual stack (a computer runs both IPv4 and IPv6 at the same time).
    - Example: Tunneling (IPv6 traffic is wrapped inside IPv4 packets so it can travel through an IPv4-only network).

## 4.2. IPv6 ADDRESSES

- Larger address space
  - IPv6: 128-bit = about  $3.4 * 10^{38}$  addresses.
    - Example: Every grain of sand on Earth could have its own IPv6 address, and we'd still have plenty left.
- Hierarchical address space
  - IPv6 addresses are structured, making routing easier.
    - Example: The beginning of an IPv6 address can show which organization or region it belongs to.
- Autoconfiguration
  - Devices can configure their own IPv6 address automatically.
    - Example: A laptop joining a Wi-Fi network can assign itself an IPv6 address without manual setup.
- Built-in Quality of Service (QoS)
  - IPv6 packets have fields to prioritize important data.
    - Example: A video call packet can be given higher priority than an email.
- Built-in security
  - IPv6 natively supports IPsec (encryption and authentication).
- Support for mobility
  - IPv6 is better at handling mobile devices that move between networks.
    - Example: A phone switching from Wi-Fi to cellular keeps its IPv6 connection smoother.

## 4.2. IPv6 ADDRESSES

### Address Structure

#### IPv6 Address Structure

- Address length
  - IPv6 = 128 bits.
  - Written as 8 groups of 16-bit hexadecimal numbers separated by colons (:).
  - Example: fe80:0:0:0:18ff:0024:8e5a:60

#### Simplification rules

- Double colon (::)
  - Consecutive groups of 0 can be replaced with ::
  - Example:
    - fe80:0:0:0:18ff:0024:8e5a:60
    - Can be written as → fe80::18ff:24:8e5a:60
  - Only one :: is allowed in an address
- Leading zeros are optional
  - 0024 → 24
  - 0060 → 60

#### Interface identifier (Windows example)

- IPv6 addresses may appear with a % number at the end.
- Example: fe80::18ff:24:8e5a:60%3
- %3 = interface index (helps Windows know which network adapter the address belongs to: Ethernet or Wi-Fi port).

#### Hexadecimal system

- Chosen because it's easier to convert to binary than decimal.
- Each hex digit = 4 bits.
- Example:
  - First field fe80 → binary: 1111 1110 1000 0000

#### Prefix length

- IPv6 uses a prefix length instead of a subnet mask.
- For host addresses, prefix is always /64  
(first 64 bits = network ID, last 64 bits = host ID).
- Example:
  - fe80::18ff:24:8e5a:60/64
  - Often, /64 is omitted because it's understood.

## 4.2. IPv6 ADDRESSES

### Address Structure

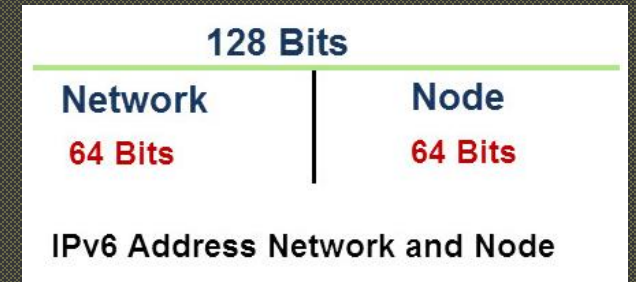
- **Network ID:** Examine the first 64 bits (equivalent to the first 16 hexadecimal digits or the first four fields separated by colons). This identifies the network portion of the address.
- **Interface ID:** Examine the last 64 bits (the remaining 16 hexadecimal digits or the last four fields). This identifies the specific device or interface on the network.
- **Example:** For an IPv6 address like 2001:0db8:85a3:0000:0000:8a2e:0370:7334
- **Network ID:** 2001:0db8:85a3:0000 (first 64 bits / four fields).
- **Interface ID:** 0000:8a2e:0370:7334 (last 64 bits / four fields).

Ways to Assign the Interface ID (host ID in IPv4) -

- Using the MAC address (EUI-64 method)
  - MAC = 48 bits → converted to 64 bits by:
    - Inserting fffe in the middle
    - Changing first two hex digits (usually replacing with 02)
  - Example:
    - MAC: 00-0C-29-7C-F9-C4
    - Interface ID: 020c:29ff:fe7c:f9c4

The 02 in the Interface ID indicates a globally unique, modified EUI-64 address (MAC 00), aligning with IPv6's requirement for a unique identifier per device on the network.

- Random permanent identifier
  - System generates a random 64-bit ID and keeps it permanently.
  - Example: Used by Windows (Vista and later) for Ethernet cards.
  - Can be changed to EUI-64 with command:
    - netsh interface ipv6 set global randomizeidentifiers=disabled
- Temporary random identifier
  - Generated randomly each time IPv6 starts.
  - Used for privacy (anonymity).
  - Example: PPP (Point-to-Point Protocol dial-up) connections.
- DHCPv6
  - Interface ID (and full address) assigned by a DHCPv6 server.
- Manual configuration
  - Administrator types the address manually in network settings (similar to IPv4).



## 4.2. IPv6 ADDRESSES

### Unicast Addresses

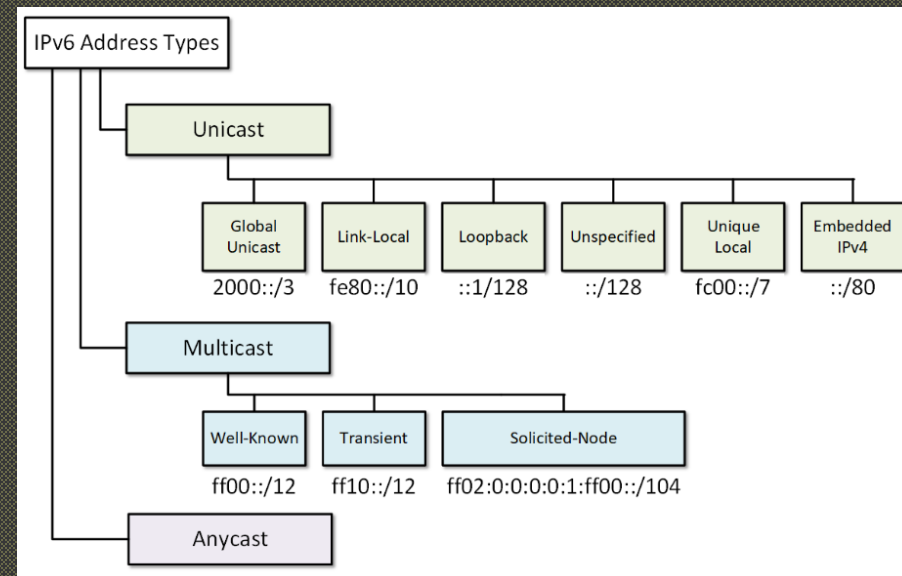
IPv6 Unicast Address (one-to-one communication): Identifies a single interface on a device.

- Requirement: Every device in IPv6 must have at least one unicast address.
  - Usually, each interface gets its own unicast address.
  - Sometimes, multiple interfaces share one unicast address (for load balancing).
- Types of Unicast Addresses:
  - Link-local
  - Unique local
  - Global
  - Plus some special and transition addresses (used for IPv4 → IPv6 migration).

- Unicast = one-to-one address.
- All IPv6 devices must have at least one.
- Link-local (fe80::) is the most basic type, automatically assigned, but only works inside the local network.

#### 1. Link-Local Address

- Always starts with: fe80
- Self-configuring → devices generate these automatically.
- Scope: Works only on the local link (not routable).
- Similar to IPv4's APIPA (169.254.x.x).
- Use case:
  - Small networks without routers.
  - Devices can talk directly to each other.
- Router rule: Routers never forward link-local traffic.



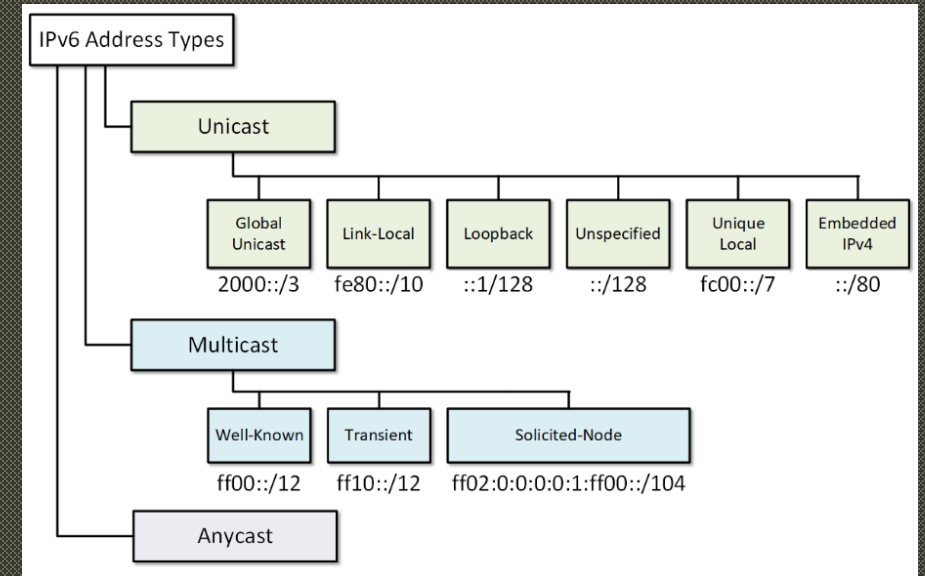
## 4.2. IPv6 ADDRESSES

### Unicast Addresses

#### 2. Unique Local IPv6 Addresses (ULA)

- Purpose
  - Similar to private IPv4 addresses (10.x.x.x, 172.16–31.x.x, 192.168.x.x).
  - Used inside private networks (not routable on the Internet).
  - Good for small/medium businesses behind a firewall.
- Prefix
  - Start with fc or fd.
  - Most common = fd (locally assigned).
  - Example prefix: fd00::/7
- Address structure
- fdgg:gggg:gggg:ssss:iiii:iiii:iiii:iiii
  - gggg... = 40-bit Global ID (unique site identifier, usually random).
  - ssss = 16-bit Subnet ID (allows up to 65,536 subnets per site).
  - iiii... = 64-bit Interface ID (specific device).

- Unique Local Addresses (fd00::/7) = IPv6's version of private IPv4 addresses.
- They are used inside private networks, offer massive subnetting, and ensure uniqueness with a pseudo-random Global ID.



<https://www.networkacademy.io/ccna/ipv6/ipv6-address-types>

#### Example

- fd12:3456:789a:0001:abcd:1234:5678:9abc
  - fd12:3456:789a → Global ID (site)
  - 0001 → Subnet ID
  - abcd:1234:5678:9abc → Interface ID

## 4.2. IPv6 ADDRESSES

### Unicast Addresses

#### 3. Global Unicast IPv6 Addresses

- Equivalent to public IPv4 addresses.
- Can be routed on the Internet and accessed globally.
- Used when a device needs direct communication across the Internet.
- Prefix
  - Start with 2 or 3 in hex (2000::/3).
  - Example: 2001:0db8:85a3:0000:0000:8a2e:0370:7334

The first 16 bits of 2000:: are 0010 0000 0000 0000 in binary, which translates to 2000 in hex. This aligns with the first three bits being 001.

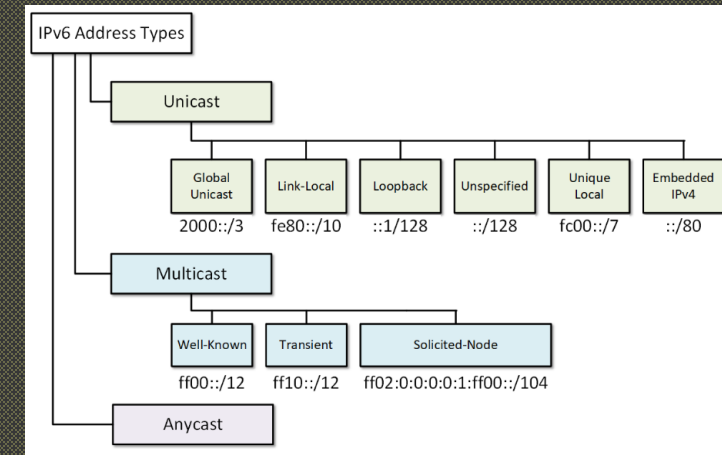
#### • Structure

- 2ggg:gggg:gggg:ssss:iiii:iiii:iiii:iiii
  - gggg... = Global ID / Global Routing Prefix (identifies a block of addresses).
  - ssss = Subnet ID (allows subnetting within an organization).
  - iiii... = Interface ID (specific device on the subnet, always 64 bits).

#### • Routing hierarchy advantage

- IPv6 global addresses are structured to make routing more efficient than IPv4.
- Example: Routers can aggregate routes by Global ID instead of storing millions of individual addresses.

- Address space
  - Global unicast = 2000::/3 → first 3 bits = 001 (2 or 3 in hex).
  - Represents 1/8 of IPv6 space (huge) –  $2^{125}/2^{128}$ .
  - Provides:
    - ~35 trillion global prefixes.
    - Each prefix → up to 65,536 subnets.
    - Each subnet → ~18 quintillion hosts.
- Flexibility
  - Normally: 48-bit Global ID + 16-bit Subnet ID + 64-bit Interface ID.
  - Can vary (larger/smaller Global ID or Subnet ID), but not common.



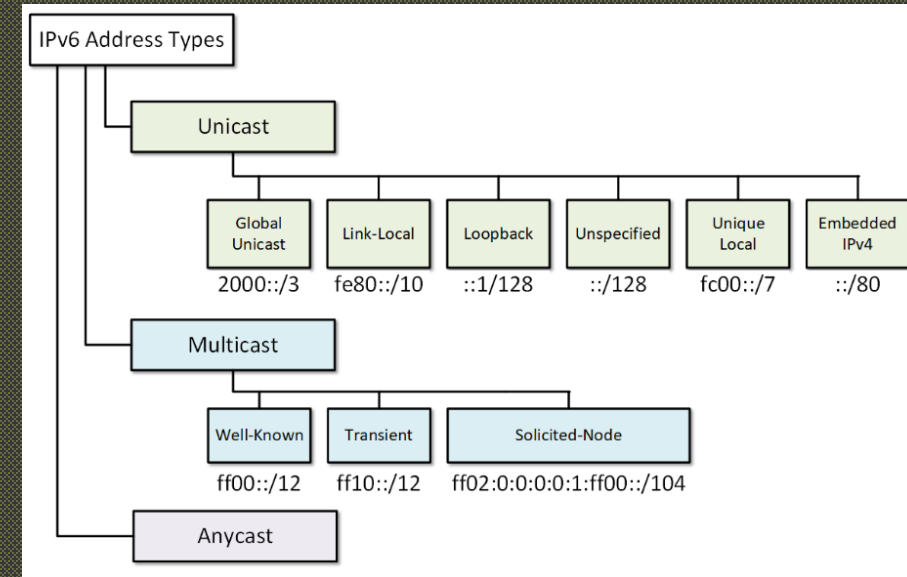
<https://www.networkacademy.io/ccna/ipv6/ipv6-address-types>

## 4.2. IPv6 ADDRESSES

### Unicast Addresses

#### 4. IPv6 Special-Purpose Addresses

- Loopback Address
  - Equivalent to IPv4 127.0.0.1.
  - Written as ::1.
  - Used for testing local IPv6 operation.
  - Example: Pinging ::1 checks if IPv6 is working on your own computer.
- Unspecified (Zero) Address
  - Written as ::.
  - Used as a placeholder when a device doesn't yet have an IPv6 address.
  - Example: Appears in the source address field before autoconfiguration finishes.
- Documentation Address
  - Prefix: 2001:db8::/32.
  - Reserved for examples, books, and training (not for real networks).
  - Example: 2001:db8:abcd:0012::1
- IPv4-to-IPv6 Transition Addresses
  - Special prefixes exist to allow IPv4 and IPv6 to work together.
  - Used for dual-stack, tunneling, and migration methods.
  - Example: ::ffff:192.168.1.1 (an IPv4-mapped IPv6 address).

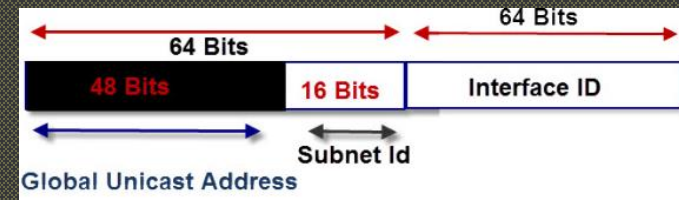


<https://www.networkacademy.io/ccna/ipv6/ipv6-address-types>

## 4.2. IPv6 ADDRESSES

### Subnetting

- Key difference from IPv4
  - IPv4: Subnetting was mainly used to save address space (scarce resource).
  - IPv6: Address space is huge → subnetting is about organization and structure, not saving addresses.
- Default ISP allocation
  - Most customers (even small home networks) get a /48 prefix.
  - /48 = first 48 bits are the network ID.
  - Remaining = 80 bits.
    - 64 bits = interface ID (host).
    - 16 bits = subnet ID.
  - → That gives 65,536 subnets for one /48 allocation.
- Example 1: No subnetting
  - A small network can leave subnet ID = 0000.
  - Example address:  
2001:db8:a00:0000:020c:29ff:fe7c:f9c4/64
  - Breakdown:
    - 2001:db8:a00 → Network prefix (/48)
    - 0000 → Subnet ID
    - Last 64 bits → Interface ID (device address)



- Example 2: Simple subnetting
  - Use the 16 subnet bits to number subnets.
  - Example:
    - 2001:db8:a00:0000::/64 → Subnet 0
    - 2001:db8:a00:0001::/64 → Subnet 1
    - 2001:db8:a00:0002::/64 → Subnet 2
- Example 3: Large organizations (structured subnetting)
  - Assign blocks of subnets to each site.
  - Example:
    - 2001:db8:a00:0000–3fff → New York (16,384 subnets)
    - 2001:db8:a00:4000–7fff → London (16,384 subnets)
    - 2001:db8:a00:8000–bfff → Shanghai (16,384 subnets)
- Why easier than IPv4?
  - Always /64 for hosts → no tricky subnet mask math.
  - Subnetting = just using the 16 subnet bits logically.

## 4.2. IPv6 ADDRESSES

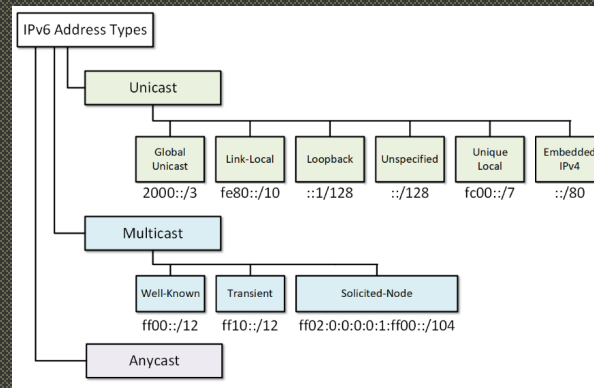
### Multicast Addresses

- Used to send one packet to many devices (a group) at once.
- Similar to IPv4 multicast, but IPv6 defines it more clearly.
- Not assigned to a single interface → instead, devices "listen" to multicast groups.
- Always begins with ff (first 8 bits = all 1s).

#### Structure of a Multicast Address

- `ffxy:zzzz:zzzz:zzzz:zzzz:zzzz:zzzz:zzzz`
- `ff` → Multicast prefix (fixed).
- `x` → Flags field (4 bits).
- `y` → Scope field (4 bits).
- `z` → Group ID (112 bits).

- Examples:
- All nodes (on local link)
  - `ff02::1` → Sent to all IPv6-enabled devices on the same link.
- All routers (on local link)
  - `ff02::2` → Sent only to routers.
- Multicast DNS (mDNS)
  - `ff02::fb` → Used for service discovery (similar to IPv4 224.0.0.251).
- Ongoing applications
  - Example: video streaming to a group of employees.
  - A group might join multicast address like `ff3e::1234` → Scope = Global (E), transient group ID = 1234.



<https://www.networkacademy.io/ccna/ipv6/ipv6-address-types>

1. Flags (x)
  - Control how the multicast address is used.
  - Bits meaning:
    - R flag = Rendezvous point (if set = 1, must also set P and T).
    - P flag = Based on network prefix.
    - T flag = Transient (1 = temporary, 0 = permanent/well-known).
  - Common values: 0, 1, 3, 7.
  - Example:
    - `ff02::1` → Flag value = 0 (well-known permanent).
2. Scope (y)
  - Defines where the multicast packets can travel.
  - Common values:
    - 1 → Interface-local (loopback, stays in same device).
    - 2 → Link-local (stays within local network, not routed).
    - 5 → Site-local (used within a site, e.g., for routers, DHCP).
    - 8 → Organization-local (stays inside the organization).
    - E → Global (can be routed across the Internet).
3. Group ID (z)
  - Identifies the specific multicast group/application.
  - 112 bits long.
  - Example: `ff02::1` → Group ID = 1 = all nodes.
  - Example: `ff02::2` → Group ID = 2 = all routers.

## 4.2. IPv6 ADDRESSES

### Anycast Addresses

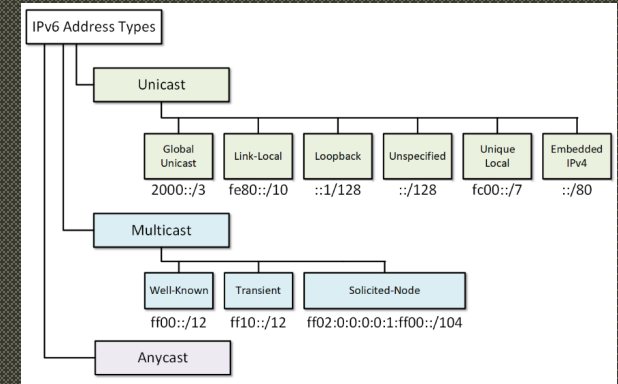
- Unique to IPv6 → no equivalent in IPv4.
- Hybrid of unicast + multicast.
- Multiple devices share the same address.
- When a packet is sent to an anycast address → it goes to the nearest device (based on routing).
- Known as “one-to-one-of-many” communication.

#### Difference from Other Address Types

- Unicast → One device, one address → packet goes to exactly one specific device.
- Multicast → One address for many devices → packet goes to all devices in the group.
- Anycast → One address for many devices → packet goes to only the closest device.

#### Why Use Anycast?

- Improves efficiency and reliability.
- Reduces latency (packets travel shortest path).
- Provides load sharing across multiple servers.



<https://www.networkacademy.io/ccna/ipv6/ipv6-address-types>

#### Examples of Anycast

- DNS Servers
  - Multiple DNS servers can share the same anycast IPv6 address.
  - A client's query goes to the nearest DNS server, not all of them.
- Routers
  - A group of routers may have the same anycast address.
  - The client's packet automatically goes to the closest router.
- CDNs (Content Delivery Networks)
  - Websites like Google, YouTube, or Cloudflare use anycast.
  - User request goes to the closest data center for faster performance.

#### Address Format

- No special prefix for anycast.
- They look like unicast addresses, but are used in an anycast way.
- Example: 2001:db8::1
- → Could be assigned to multiple DNS servers worldwide.  
→ A client in Toronto connects to the Canadian DNS server, while a client in London connects to the UK server.

## 4.2. IPv6 ADDRESSES

### Autoconfiguration Methods

#### 1. Stateless Autoconfiguration (SLAAC)

- Device listens for Router Advertisement (RA) messages from a local router.
- If the Autonomous flag = ON → device uses the prefix in the RA + its own 64-bit interface ID to build its IPv6 address.
- If the Autonomous flag = OFF → device ignores the prefix and tries:
  - DHCPv6 (for full address info)
  - or uses its link-local address (fe80::/10).
- Example:
  - Router sends prefix: 2001:db8:1234::/64
  - Device interface ID: 020c:29ff:fe7c:f9c4
  - Final address: 2001:db8:1234::020c:29ff:fe7c:f9c4

#### 2. Stateful Autoconfiguration

- Device fully relies on DHCPv6 to get:
  - IPv6 address
  - Gateway
  - DNS info
- Used when no router is present OR router advertisement flag doesn't allow stateless config.
- Example:
  - DHCPv6 server assigns: 2001:db8:5678::25/64 + DNS info.

## 4.2. IPv6 ADDRESSES

### Autoconfiguration Methods

#### IPv6 Autoconfiguration on Windows (Step by Step)

- Generate a link-local address (always starts with fe80::/10).
- Check uniqueness using Duplicate Address Detection (DAD).
  - If duplicate → create a new one, repeat check.
- Assign link-local address if unique.
- Send Router Solicitation (RS) message → asking if any router is available.
- If no Router Advertisement (RA) received → try DHCPv6.
- If RA is received:
  - If Autonomous flag = ON → build IPv6 address using RA prefix + interface ID.
  - If Autonomous flag = OFF → use DHCPv6 to get the full address.
  - Can still use DHCPv6 for extra info (like DNS).

- Stateless = device makes its own address (prefix from router + interface ID).
- Stateful = DHCPv6 server hands out the full address.
- Windows always starts with link-local, checks uniqueness, then tries RA or DHCPv6.

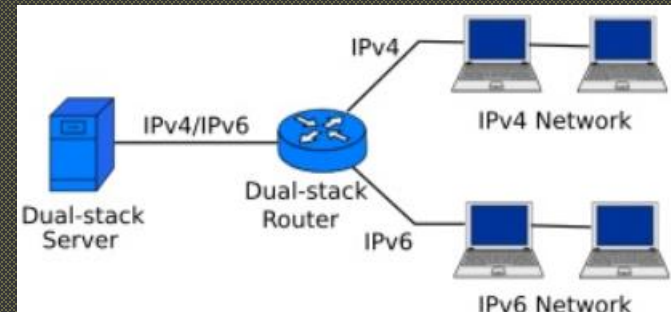
## 4.2. IPv6 ADDRESSES

### Transitioning from IPv4 to IPv6

- IPv4 is running out of addresses.
- IPv6 solves the address space problem and adds features like auto-configuration, QoS, and security.
- No set date: Transition happens gradually over years.
- Both can coexist: A computer or network can run both IPv4 and IPv6 at the same time (called dual-stack).
- Transition technologies help devices using IPv4 and IPv6 talk to each other.

#### 1. Dual IP Layer Architecture (Dual-Stack)

- Both IPv4 and IPv6 run together on the same device.
- Device can communicate with both IPv4-only and IPv6-only systems.
- Sometimes, IPv6 packets are encapsulated in IPv4 headers (tunneling).
- Example:
  - A Windows 11 PC can have:
    - IPv4: 192.168.1.10
    - IPv6: 2001:db8:abcd::10
  - It can talk to both IPv4 and IPv6 hosts.



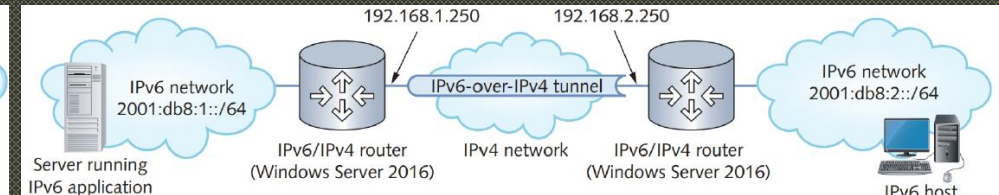
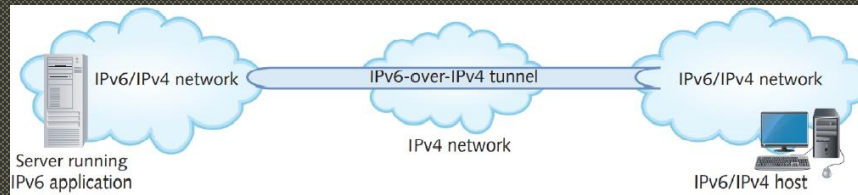
<http://bucarotechelp.com/networking/standards/80002701.asp>

## 4.2. IPv6 ADDRESSES

### Transitioning from IPv4 to IPv6

#### 2. IPv6-over-IPv4 Tunneling

- Technique to send IPv6 packets through an IPv4 network by wrapping them inside IPv4 headers.
- Needed if ISP or routers only support IPv4.
- Example:
  - IPv6 app on your PC → ISP uses IPv4 routers → packet is tunneled inside IPv4 → at destination, it's converted back to IPv6.



#### 3. ISATAP (Intra-Site Automatic Tunnel Addressing Protocol)

- Automatic tunneling protocol for IPv6 traffic inside a company's IPv4 network.
- No manual setup needed (Windows enables it by default).
- IPv6 address format includes the IPv4 address.
- Example:
  - IPv4 address: 172.31.210.200
  - ISATAP IPv6 address: fe80::5efe:172.31.210.200



## 4.2. IPv6 ADDRESSES

### Transitioning from IPv4 to IPv6

1. Dual-Stack (dual IP layer) = both, IPv4 + IPv6 run together.
2. Tunneling = wrap IPv6 inside IPv4 to travel across IPv4-only networks.
3. ISATAP = automatic tunneling for internal networks (uses fe80::5efe).
4. 6to4 = tunneling using prefix 2002::/16, needs public IPv4.
5. Teredo = tunneling using prefix 2001::/32, works behind NAT routers.

#### 4. 6to4 Tunneling

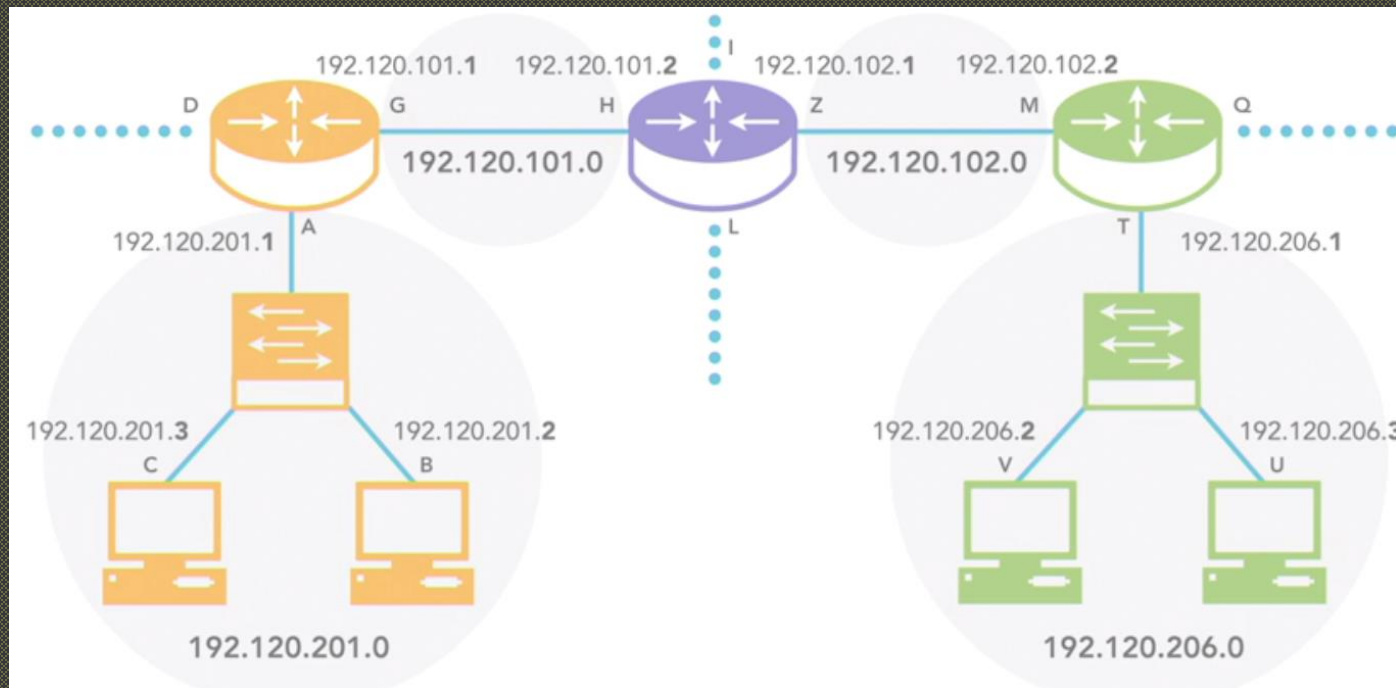
- Automatic tunneling of IPv6 traffic over IPv4 networks.
- Recognized by prefix 2002::/16.
- IPv4 address is embedded in the IPv6 address.
- Limitation: Needs a public IPv4 address (won't work with NAT).
- Example:
  - IPv4 address: 192.0.2.4
  - Hexadecimal: C000:0204
  - 6to4 IPv6 address: 2002:C000:0204::/48

#### 5. Teredo Tunneling

- Solves 6to4's limitation (works behind NAT routers).
- Encapsulates IPv6 packets inside IPv4 UDP messages, so they pass through NAT.
- Recognized by prefix 2001::/32.
- Address structure includes:
  - Teredo server IPv4 address
  - Flags
  - Obscured UDP port
  - Obscured IPv4 client address
- Example:
  - Teredo address: 2001:0000:4136:e378:8000:63bf:3fff:fdd2

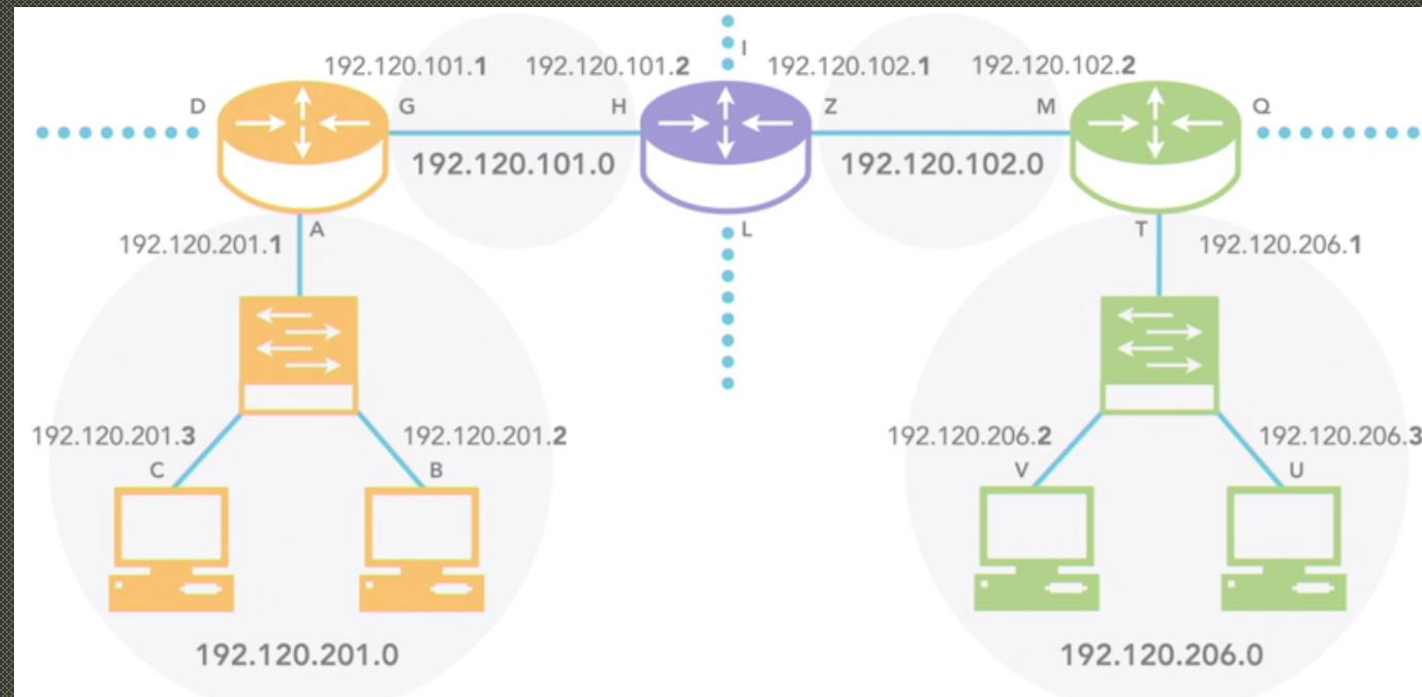
## 4.3 Digital Communications based on Physical and Logical Addresses

- Routers are presented in cylinder shape and switches in prism shape.
- A, B, C, D, G, H, Z, M, Q, T, V, and U represent MAC addresses of each device
- Each segment has the last octet of the network IP address zero
- Each of the networks (segments), that has the last octet of the IP address zero, is a subnet of the large network 192.120.0.0
- Each router interface of every subnet has a host IP address



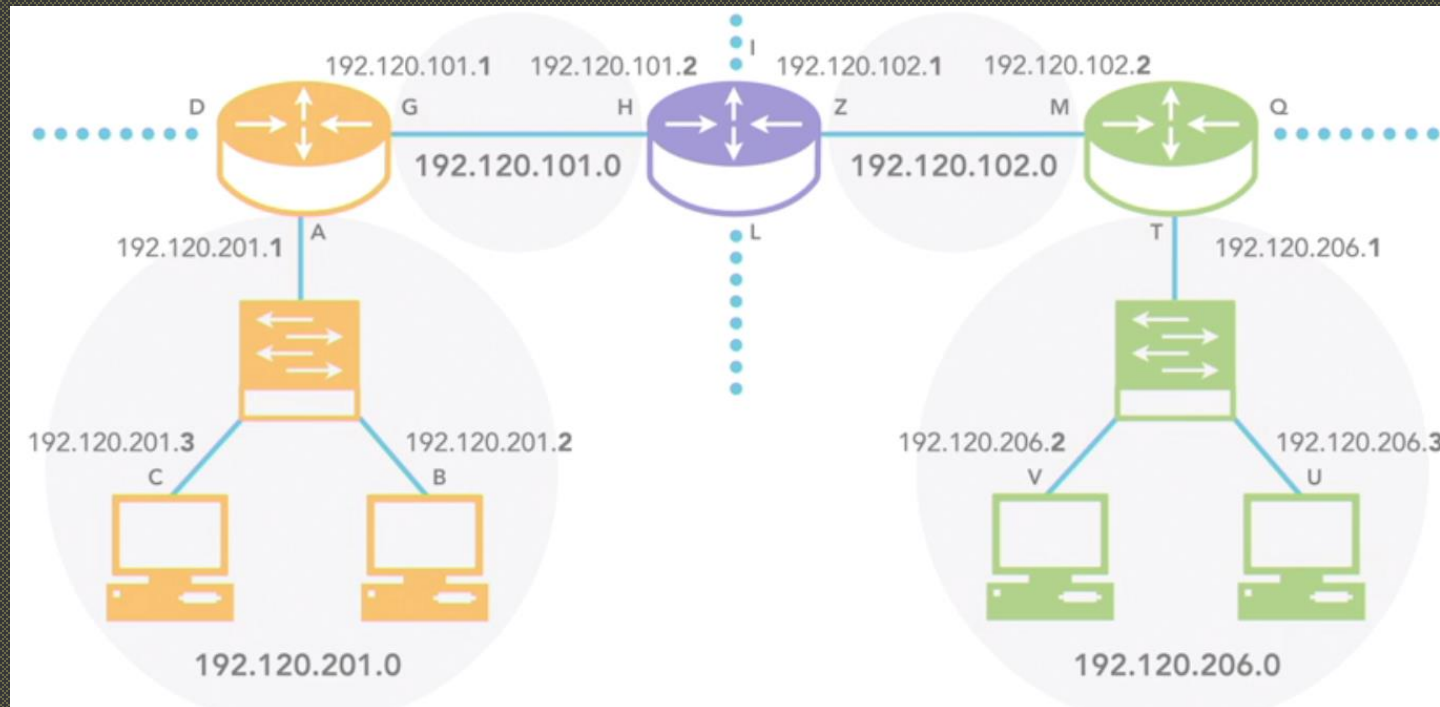
## 4.3 Digital Communications based on Physical and Logical Addresses

- Data has to move from station C to station U.
- DNS server provides the IP addresses of the source (C) and the destination (U).
- Source (C) sends the packet to the interface A of router ADG by setting the source and destination MAC addresses of C and A respectively.
- Also, for source and destination IP address puts its own (C) and that of station (U) respectively.



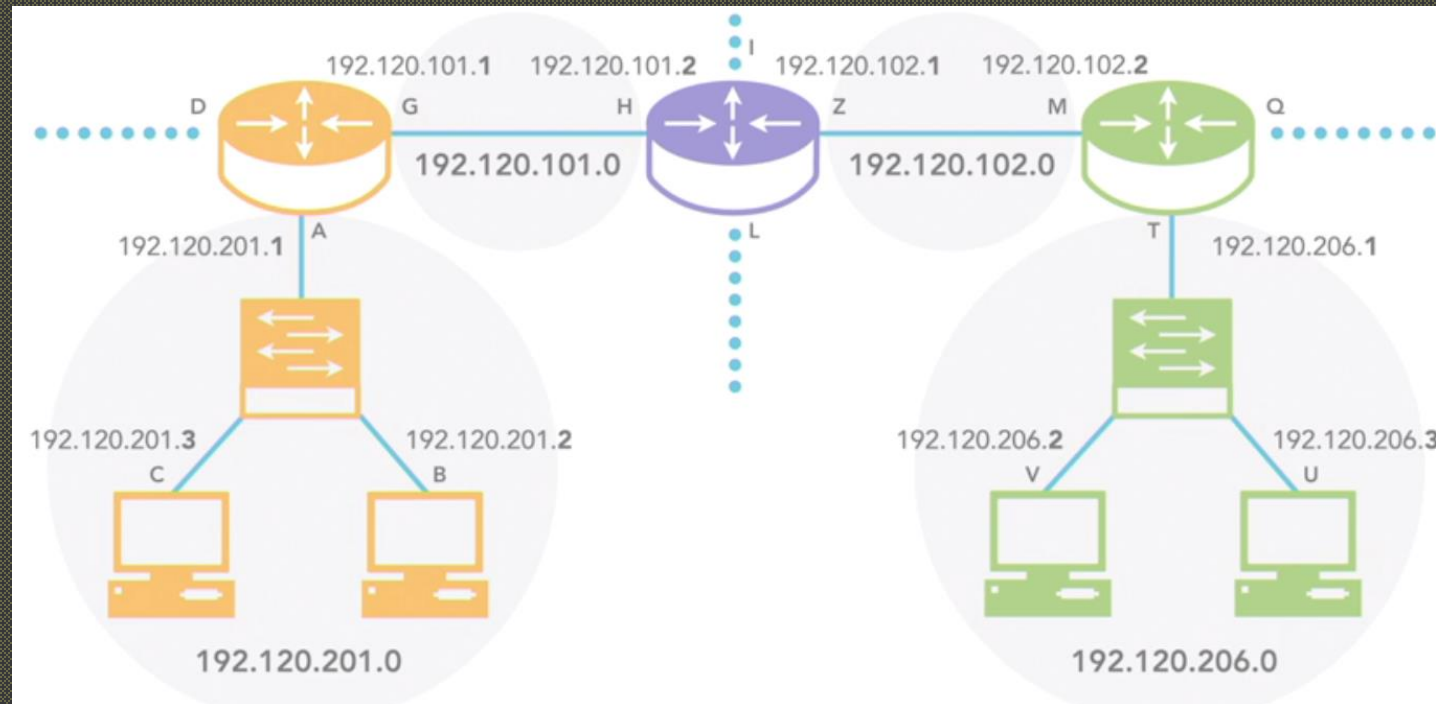
## 4.3 Digital Communications based on Physical and Logical Addresses

- The first hop is when the packet reaches router ADG which removes the MAC addresses from packet.
- Also, based on the destination IP address, checks its routing table to find out that router HZL is the one that will move the packet toward the destination (U).
- Router ADG sends the packet to the interface H of router HZL by setting the source and destination MAC addresses to G and H respectively. Also, it leaves source and destination IP address unchanged.



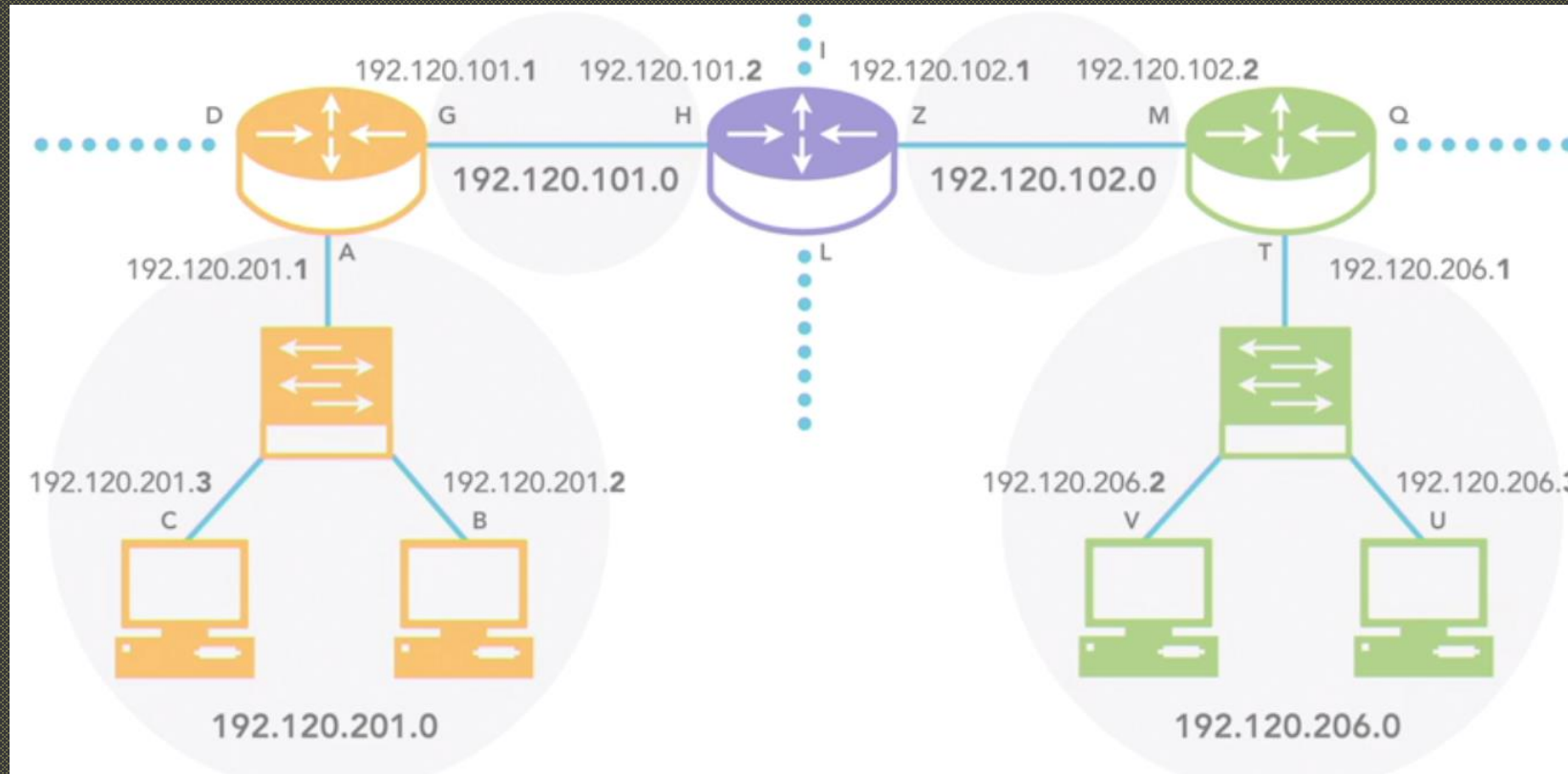
## 4.3 Digital Communications based on Physical and Logical Addresses

- The second hop is when the packet reaches router HZL which removes the MAC addresses from packet.
- Also, based on the destination IP address, checks its routing table to find out that router MQT is the one that will move the packet to the destination.
- Router HZL sends the packet to the interface M of router MQT by setting the source and destination MAC addresses to Z and M respectively. Also, it leaves source and destination IP address unchanged.



## 4.3 Digital Communications based on Physical and Logical Addresses

- Based on the IP address of station U, Router MQT will direct the packets into the switch connected to its port T.
- The switch will direct the packets into the station U based on its MAC address (U).



# References

- Guide to Networking Essentials, 8<sup>th</sup> Edition, Greg Tomsho

# INDUSTRIAL NETWORKING

## MENG 4010

Thank you!

Discussions?